RBA vs Machine: Can Algorithms Outforecast the Central Bank?

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Abstract

This paper evaluates whether machine learning methods trained on publicly available macroeconomic data can produce more accurate forecasts than the Reserve Bank of Australia's published projections. We construct a quarterly panel of 487 Australian and international time series spanning 1990 to 2025 and compare regularised linear and tree-based models against RBA forecasts for headline and underlying inflation, GDP growth, and the unemployment rate at horizons of zero to eight quarters ahead, using an expanding-window pseudo-real-time design aligned to RBA forecast origins from 1993 Q1. We find that the RBA holds a clear forecasting advantage for headline CPI inflation at short horizons, consistent with its access to non-public information and high-frequency data. At longer horizons of one to two years, however, tree-based machine learning models outperform the RBA for both inflation measures. For GDP growth, simple benchmarks match or outperform the RBA at all horizons beyond the nowcast. Diebold–Mariano tests confirm statistical significance in a limited number of cases, reflecting the modest sample sizes typical of quarterly macroeconomic data. All methods deteriorate substantially during the post-COVID period. These results support the integration of machine learning methods as a cross-check within central bank forecasting frameworks.

AI Disclosure. This paper was prepared with the assistance of artificial intelligence tools (Claude, Anthropic) for code development, data pipeline construction, and manuscript drafting. All analysis, interpretation, and conclusions are the responsibility of the authors. The replication code is available at https://github.com/andybridger/RBAvsMachine.

1 Introduction

Australia's monetary policy framework rests on macroeconomic forecasts that the Reserve Bank of Australia (RBA) publishes each quarter, yet the accuracy of these forecasts has received surprisingly little systematic evaluation. The RBA's projections for inflation, GDP growth, and unemployment—published in the *Statement on Monetary Policy*—inform interest rate decisions, shape market expectations, and serve as inputs to fiscal planning.[1] Whether these projections are accurate, and whether alternative methods could improve upon them, is a question of direct policy relevance.

This paper asks whether machine learning methods, constrained to use only publicly available data, can produce more accurate forecasts than the RBA's published projections. We focus on four target variables central to the RBA's mandate: year-ended CPI inflation, year-ended underlying (trimmed mean) inflation, quarterly GDP growth, and the unemployment rate. Forecasts are evaluated at horizons of 0 (nowcast) to 8 quarters ahead, matching the horizons reported in the *Statement on Monetary Policy*.

The question is motivated by three developments. First, the 2023 independent Review of the Reserve Bank of Australia recommended that the Bank strengthen its analytical toolkit and improve its communication of forecast uncertainty (RBA Review Panel, 2023). Second, Tulip & Wallace (2012), in the most comprehensive assessment of RBA forecast accuracy to date, found that the Bank's inflation forecasts had substantial explanatory power but that its GDP growth forecasts added little information beyond the historical mean. To our

[1] The *Statement on Monetary Policy* has been published quarterly since February 2007; prior to this it was published semi-annually. The format of the RBA's forecast tables has evolved over time, and the 2023 Review recommended further changes to the Bank's communication practices. Appendix D provides a brief history of RBA forecast publication.

knowledge, no study has updated this full cross-variable, cross-horizon evaluation. Third, the post-pandemic period—in which the RBA underestimated inflation by several percentage points within a single year (Reserve Bank of Australia, 2022)—has renewed interest in the comparative accuracy of alternative forecasting methods.

We contribute to this literature in three ways. First, we provide an updated assessment of RBA forecast performance extending through 2025, covering the pandemic, the inflationary surge, and the subsequent tightening cycle. Second, we offer the first systematic comparison of published RBA forecasts against machine learning methods using a common information set in the Australian context. Third, we draw out the implications of a central finding: the RBA holds a clear forecasting advantage at short horizons, but machine learning models—particularly tree-based methods—outperform at horizons of one to two years. We argue that this pattern is consistent with the RBA possessing superior short-run information through its liaison programme, while its longer-horizon forecasts may be anchored toward the inflation target rather than reflecting data-driven trends.

Our approach builds on a growing international literature applying machine learning to macroeconomic forecasting. Medeiros et al. (2021) demonstrate the benefits of machine learning methods for US inflation forecasting. Jung et al. (2018) and Hauzenberger et al. (2022) provide evidence from the IMF and the Bank of England, respectively. On the data side, McCracken & Ng (2016) introduced the FRED-MD monthly database, which has become a standard benchmark for US forecasting studies. McCracken & Ng (2021) extended this to quarterly frequency. Similar databases have been constructed for Canada (Fortin-Gagnon et al., 2022) and Australia (Panagiotelis et al., 2019; Tsiaplias & Chua, 2010). We follow the conventions established in this literature regarding data transformations, stationarity testing, and pseudo-real-time evaluation design.

The remainder of the paper proceeds as follows. Section 2 describes the data. Section 3 presents the forecasting methods and evaluation metrics. Section 4 reports results. Section 5 discusses robustness. Section 6 interprets the findings and discusses limitations. Section 7 concludes.

2 Data

2.1 Sources

The macroeconomic panel is assembled from four primary sources, chosen to approximate the breadth of information available to the RBA's forecasting team at each forecast origin.

Australian Bureau of Statistics (ABS). We obtain 319 time series from the ABS using the `readabs` R package (Cowgill, 2023a). These include the Consumer Price Index and expenditure-class components, trimmed mean and weighted median measures of underlying inflation, real GDP and its expenditure components, labour force statistics, wage price indices, and building approvals.

Reserve Bank of Australia (RBA). An additional 153 series are obtained via the `readrba` package (Cowgill, 2023b), including inflation expectations, financial market variables (the cash rate, yield curve, government and corporate bond yields), credit and lending aggregates, exchange rates, commodity price indices, and the RBA's historical forecast tables. The historical forecast data are constructed following the methodology of Tulip & Wallace (2012).

Federal Reserve Economic Data (FRED). Five international commodity price series are drawn from FRED: Brent crude oil, West Texas Intermediate crude oil, iron ore, Australian thermal coal, and a global Brent crude measure.

Yahoo Finance. Ten equity market indices are included: the ASX 200, All Ordinaries, S&P 500, FTSE 100, Nikkei 225, KOSPI, Hang Seng, Shanghai SSE Composite, Euronext 100, and gold. Daily data are aggregated to quarterly frequency by averaging within each quarter.

The combined panel contains 487 unique time series observed at monthly and quarterly frequencies through December 2025. Appendix A provides a summary of variable categories, and Appendix B lists all series with their identifiers and transformation codes. All data retrieval is fully reproducible through the project's R pipeline.

2.2 Frequency Conversion

Series observed at monthly frequency are converted to quarterly frequency prior to modelling. For stock variables and rates (such as the unemployment rate, interest rates, and price index levels), the quarterly value is the average of the three monthly observations within the quarter. Financial market data from Yahoo Finance, observed at daily frequency, are converted by averaging daily closing prices within each quarter. These conventions follow McCracken & Ng (2016). The frequency of each series and its source are reported in Appendix B.

2.3 Transformations

Following the conventions established by McCracken & Ng (2016) for the FRED-MD database, we apply one of seven transformation codes to each series to induce approximate stationarity.

The transformation for each series is selected by augmented Dickey–Fuller tests applied to the level, first difference, and log first difference. Table 1 summarises the transformation codes.

Table 1: Transformation codes applied to induce stationarity.

Code	Transformation	Description
1	x_t	No transformation (level)
2	Δx_t	First difference
3	$\Delta^2 x_t$	Second difference
4	$\ln x_t$	Log level
5	$\Delta \ln x_t$	Log first difference (growth rate)
6	$\Delta^2 \ln x_t$	Log second difference
7	$\Delta(x_t/x_{t-1} - 1)$	Percent change difference

The majority of series (360 of 487) are rendered stationary by log first differencing (code 5), consistent with the finding in McCracken & Ng (2016) that most macroeconomic aggregates are well characterised as integrated of order one in logs. Series containing zeros or negative values are differenced rather than log-differenced. The complete mapping of series to transformation codes is provided in Appendix B.

2.4 Target Variables and Forecast Horizons

We evaluate forecasts for four target variables:

- **CPI inflation**: year-ended percentage change in the headline Consumer Price Index (quarterly).

- **Underlying inflation**: year-ended trimmed mean inflation (quarterly), the RBA's preferred measure of underlying price pressure.
- **GDP growth**: quarter-on-quarter percentage change in real chain-volume GDP. We note that GDP growth is not a direct target of the RBA's mandate, which focuses on price stability and maximum sustainable employment; we include it for completeness and comparability with Tulip & Wallace (2012).
- **Unemployment rate**: the seasonally adjusted unemployment rate, converted from monthly to quarterly frequency by averaging.

Forecasts are evaluated at horizons of 0 (nowcast), 1, 2, 4, and 8 quarters ahead, corresponding to the horizons reported in the *Statement on Monetary Policy*. Following Tulip & Wallace (2012), we restrict the evaluation sample to forecasts made from 1993 Q1 onward, when the RBA formally adopted its 2–3 per cent inflation target.

2.5 RBA Forecast Errors

RBA forecasts are extracted from the `readrba` package's `read_forecasts()` function. Forecast errors are defined as

$$e_{t+h}^{\mathrm{RBA}} = y_{t+h} - \hat{y}_{t+h|t}^{\mathrm{RBA}}, \tag{1}$$

where y_{t+h} is the realised value and $\hat{y}_{t+h|t}^{\mathrm{RBA}}$ is the RBA's forecast made at time t for period $t+h$. Realised values are taken from the latest available release of the data, following Tulip & Wallace (2012). This introduces a modest advantage for the RBA relative to a pure real-time evaluation, since some early forecasts were assessed against data that have subsequently been revised. The direction of this bias is ambiguous: data revisions may either increase or decrease the apparent accuracy of the original forecast, depending on whether the revision moves the realised value toward or away from the initial projection.

3 Machine Learning Methods

3.1 Model Suite

We employ seven forecasting methods, spanning naive benchmarks, regularised linear models, and nonlinear ensemble methods.

Historical mean. The forecast is the sample average of the target variable up to the forecast origin. This serves as the most parsimonious benchmark: any method that cannot improve upon the historical mean adds noise rather than signal.

AR(1). A first-order autoregressive model estimated by ordinary least squares on the target variable alone. This captures persistence without using the predictor panel.

Ridge regression (Hoerl & Kennard, 1970). Linear regression with an ℓ_2 penalty. Ridge shrinks coefficients toward zero without setting any exactly to zero, making it suitable when many predictors carry small amounts of information.

LASSO (Tibshirani, 1996). Linear regression with an ℓ_1 penalty. The LASSO performs simultaneous estimation and variable selection, setting irrelevant coefficients exactly to zero.

Elastic net (Zou & Hastie, 2005). A convex combination of the ℓ_1 and ℓ_2 penalties ($\alpha = 0.5$), combining the variable-selection properties of the LASSO with the grouping behaviour of Ridge.

Random Forest (Breiman, 2001). An ensemble of regression trees, each trained on a bootstrap sample with a random subset of predictors at each split. Random Forests capture nonlinearities and interactions without parametric assumptions.

XGBoost (Chen & Guestrin, 2016). Extreme gradient boosting builds an ensemble of trees sequentially, with each tree fitting the residuals of the previous ensemble. We use moderate regularisation (maximum depth 3, learning rate 0.1, subsample and column-sample ratios of 0.8) and 20 boosting rounds.

3.2 Estimation Procedure

All models are estimated using a direct forecasting approach with an expanding window that respects the real-time information constraint. For each RBA forecast origin at time t and horizon h, the model estimates

$$y_{t+h} = f(\mathbf{X}_t) + \varepsilon_{t+h}, \tag{2}$$

where $\mathbf{X}_t$ is the vector of predictors available at time t. The estimation proceeds as follows:

1. Construct the predictor panel using all observations available up to quarter t.
2. Align the target variable y_{t+h} with the predictors $\mathbf{X}_t$ so that the regression estimates $E[y_{t+h} \mid \mathbf{X}_t]$ directly, following Stock & Watson (2002).
3. Standardise all predictors to zero mean and unit variance using only training-sample

moments.

4. For regularised linear models, select the penalty parameter via BIC-based selection along the `glmnet` regularisation path, following Hastie et al. (2009).

5. Generate the forecast $\hat{y}^m_{t+h|t}$ and record the forecast error $e^m_{t+h} = y_{t+h} - \hat{y}^m_{t+h|t}$.

The minimum training sample is set at 40 quarterly observations. Columns with zero variance are dropped, and remaining missing values are set to zero after standardisation. These choices follow standard practice in the large-dimensional forecasting literature (McCracken & Ng, 2016; Stock & Watson, 2002).

3.3 Evaluation Metrics

Forecast accuracy is assessed using three metrics. The root mean squared forecast error (RMSFE) is defined as

$$\mathrm{RMSFE}^m_h = \sqrt{\frac{1}{T_h}\sum_{t=1}^{T_h}\left(e^m_{t+h}\right)^2}, \tag{3}$$

the mean absolute forecast error (MAFE) as

$$\mathrm{MAFE}^m_h = \frac{1}{T_h}\sum_{t=1}^{T_h}\left|e^m_{t+h}\right|, \tag{4}$$

and the forecast bias (mean error) as

$$\mathrm{Bias}^m_h = \frac{1}{T_h}\sum_{t=1}^{T_h} e^m_{t+h}. \tag{5}$$

We report the relative RMSFE of each model against the RBA benchmark,

$$\text{Relative RMSFE}_h^m = \frac{\text{RMSFE}_h^m}{\text{RMSFE}_h^{\text{RBA}}}, \tag{6}$$

where values below one indicate that model m outperforms the RBA at horizon h.

Statistical significance of forecast accuracy differences is assessed using the Diebold & Mariano (1995) test. Define the loss differential $d_t = (e_{t+h}^{\text{RBA}})^2 - (e_{t+h}^m)^2$. The test statistic is

$$\text{DM} = \frac{\bar{d}}{\sqrt{\widehat{\text{Var}}(\bar{d})/T_h}}, \tag{7}$$

where $\widehat{\text{Var}}(\bar{d})$ is a heteroskedasticity and autocorrelation consistent (HAC) variance estimator using $h-1$ autocovariances. The bandwidth $h-1$ follows from the fact that optimal direct multi-step forecasts are at most MA($h-1$) (Diebold & Mariano, 1995). Positive values of DM indicate that model m outperforms the RBA. Full RMSFE, MAFE, bias, and DM test results are reported in Appendix C.

4 Results

4.1 RBA Forecast Performance

Before comparing the RBA to machine learning methods, we characterise the RBA's forecast accuracy over the evaluation sample. The density of RBA forecast errors across horizons (Figure 1) reveals several patterns. For CPI and underlying inflation, the distributions are approximately centred at zero for short horizons but develop positive skewness at longer horizons, reflecting the upward inflation surprises of 2021–2023. The unemployment rate forecasts show a symmetric pattern at short horizons but tend toward overprediction at longer horizons.

For GDP growth, the forecast error distributions are wide even at short horizons, consistent with Tulip & Wallace (2012)'s finding that GDP growth is inherently difficult to predict. The RBA's RMSFE for GDP growth exceeds 1.0 percentage points at all horizons beyond the nowcast.

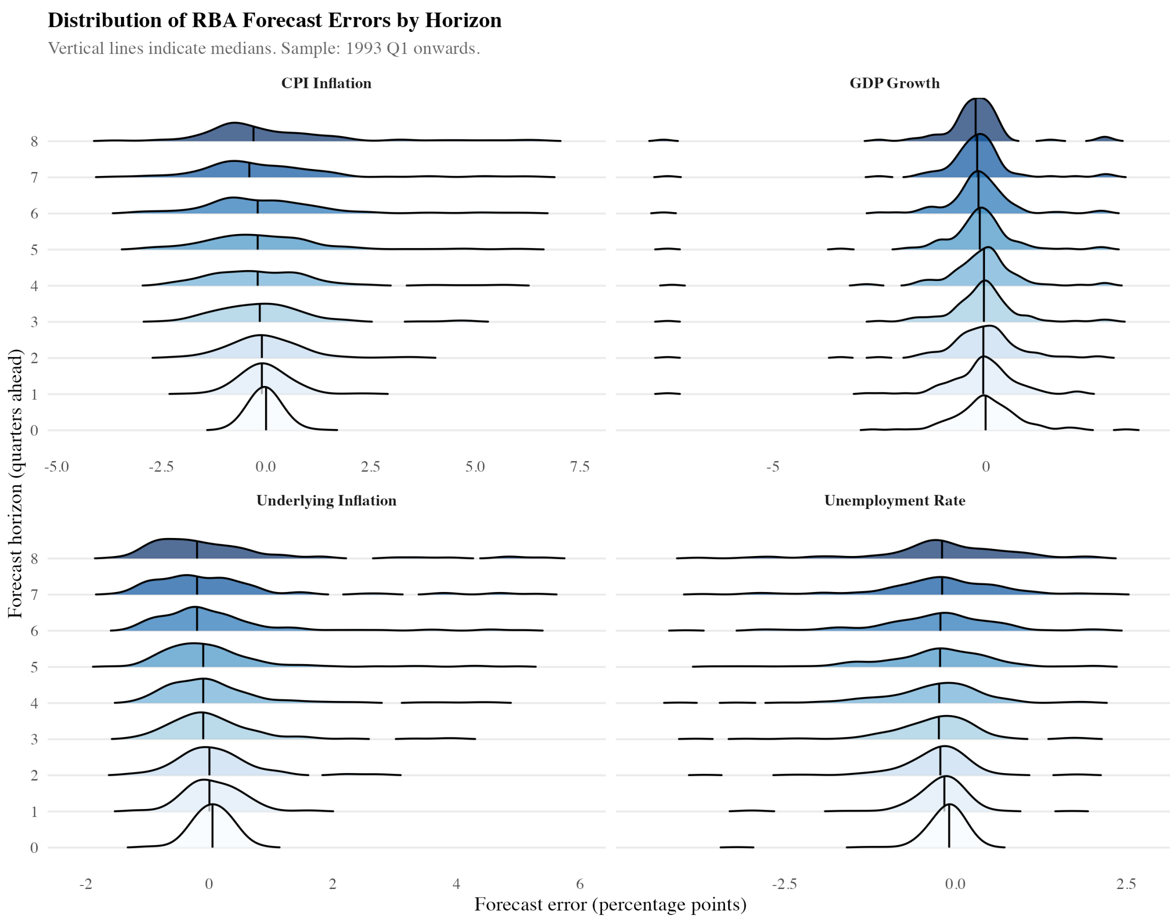

Figure 1: Distribution of RBA forecast errors by horizon. Vertical lines indicate medians. Sample: 1993 Q1 onwards.

Figure 2 presents forecast paths against realised values for CPI inflation and the unemployment rate since 2015. The COVID-19 period is visible as a cluster of large forecast revisions, followed by systematic underprediction of inflation and overprediction of unemployment during the recovery. The largest errors are concentrated in a small number of episodes rather than being uniformly distributed across the sample.

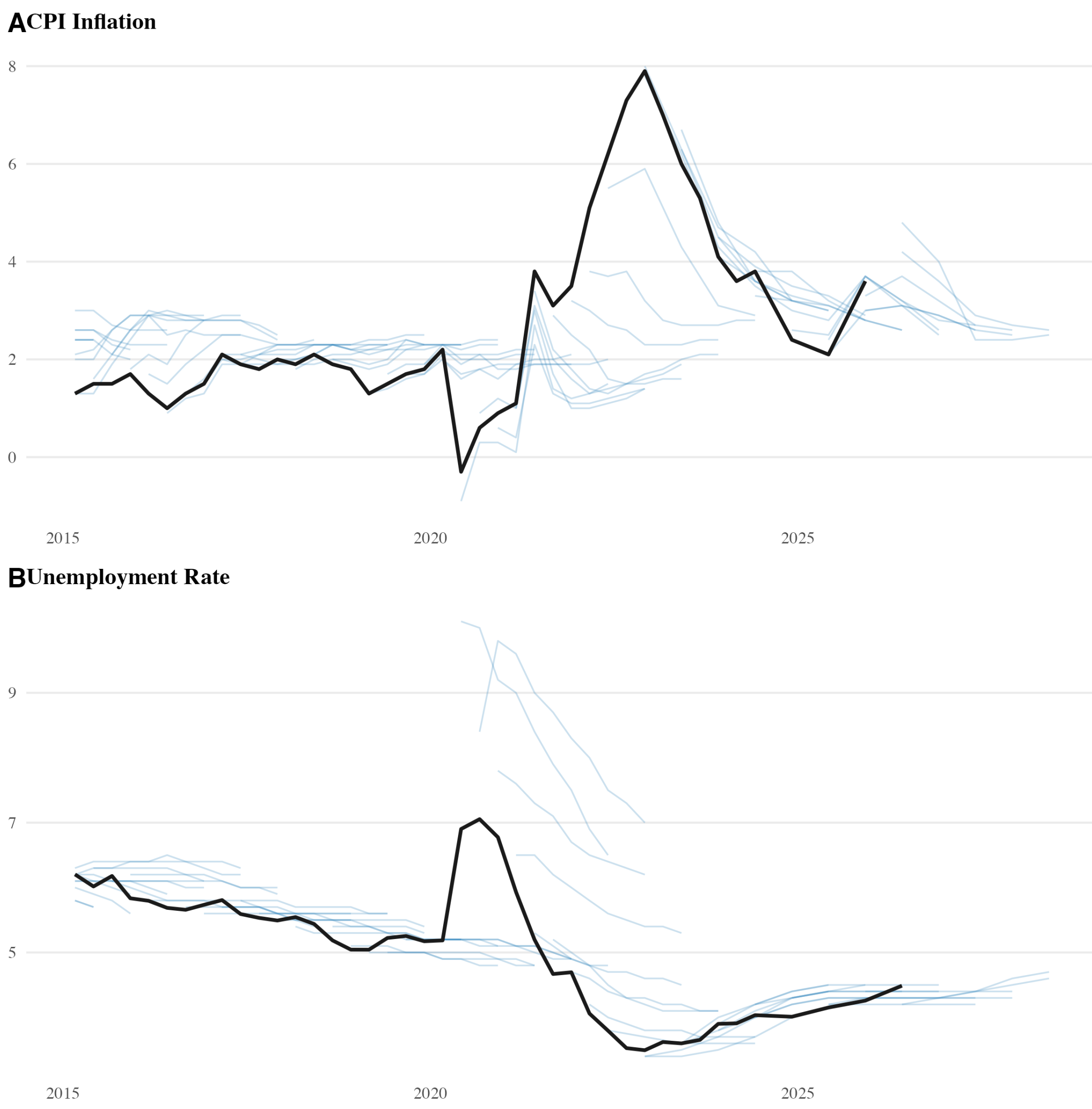

Figure 2: RBA forecast paths (blue) versus realised values (black) for CPI inflation and the unemployment rate, 2015–2025.

4.2 Main Results

Table 2 in Appendix C reports the full RMSFE of each model at all forecast horizons. Figure 3 displays the relative RMSFE of each machine learning model against the RBA benchmark. The central finding is that the relative performance of machine learning and the RBA depends systematically on the forecast horizon. We discuss each target variable in turn.

CPI inflation. At short horizons (one to two quarters ahead), the RBA outperforms every model in the panel. The best-performing alternative at $h = 1$ is the AR(1), with a relative RMSFE of 1.34—meaning it is 34 per cent less accurate than the RBA. At the four-quarter horizon, the gap narrows: Random Forest achieves a relative RMSFE of 0.99. At $h = 8$ (two years ahead), tree-based methods outperform the RBA, with Random Forest achieving a relative RMSFE of 0.79 and XGBoost 0.89. Diebold–Mariano tests, however, do not reject the null of equal predictive accuracy at $h = 8$ (DM = 0.80, $p = 0.43$ for Random Forest), reflecting the small evaluation sample of 72 observations at that horizon.

Underlying inflation. For trimmed mean inflation, the AR(1)—which simply extrapolates recent persistence—outperforms the RBA at every horizon. At $h = 1$, the AR(1) achieves a relative RMSFE of 0.75, and the Diebold–Mariano test confirms this is statistically significant (DM = 4.05, $p < 0.001$). At $h = 8$, Random Forest achieves a relative RMSFE of 0.66, a 34 per cent improvement over the RBA. The strong performance of the AR(1) reflects the high persistence of underlying inflation: trimmed mean inflation, by construction, removes volatile items, leaving a series dominated by slow-moving trends that autoregressive models capture well.

GDP growth. At the nowcast horizon ($h = 0$), machine learning methods achieve dramatic improvements: the Elastic Net and LASSO produce RMSFEs of 0.11–0.12, compared to the RBA's 0.80. This reflects the fact that many quarterly indicators have already been released by the time of the nowcast, and regularised methods can extract this information efficiently. At horizons of one quarter and beyond, however, the LASSO and Elastic Net converge to the historical mean, selecting no predictors from the panel. This result implies that no publicly available predictor in the 487-variable panel systematically helps forecast GDP growth at horizons beyond the current quarter. The RBA's RMSFE for GDP growth at $h = 8$ is 1.20, while simple benchmarks achieve approximately 0.98 (relative RMSFE 0.82). Diebold–Mariano tests do not reject equal accuracy at any horizon beyond $h = 0$.

Unemployment rate. The AR(1) is substantially more accurate than the RBA at short horizons, achieving a relative RMSFE of 0.51 at $h = 1$ (DM = 2.21, $p = 0.027$). This advantage diminishes at longer horizons: at $h = 8$, the AR(1) relative RMSFE is 1.00, essentially identical to the RBA. Unlike the inflation variables, more complex machine learning methods (Random Forest, XGBoost) do not improve upon the AR(1) for unemployment at any horizon.

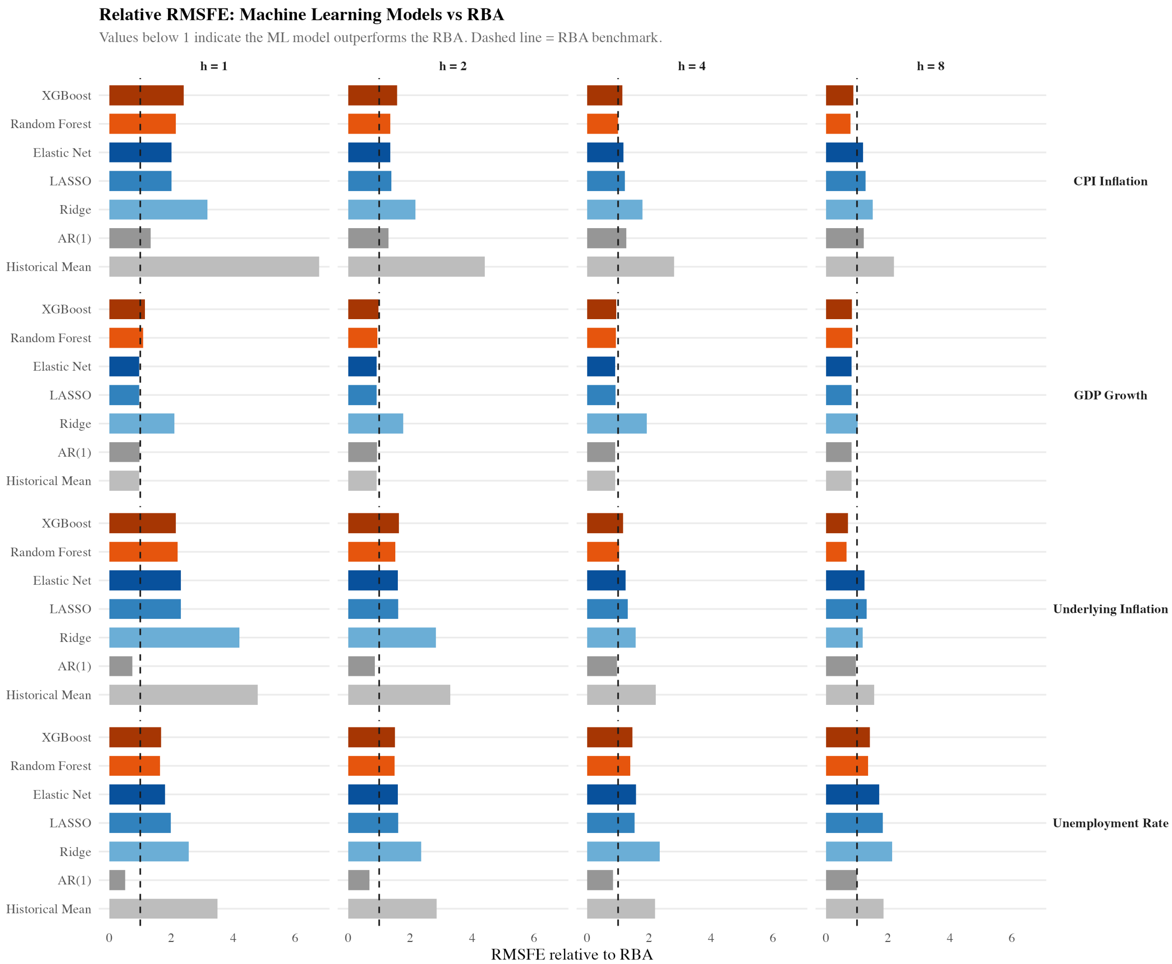

Figure 3: Relative RMSFE of machine learning models against the RBA. Values below the dashed line indicate that the ML model outperforms the RBA.

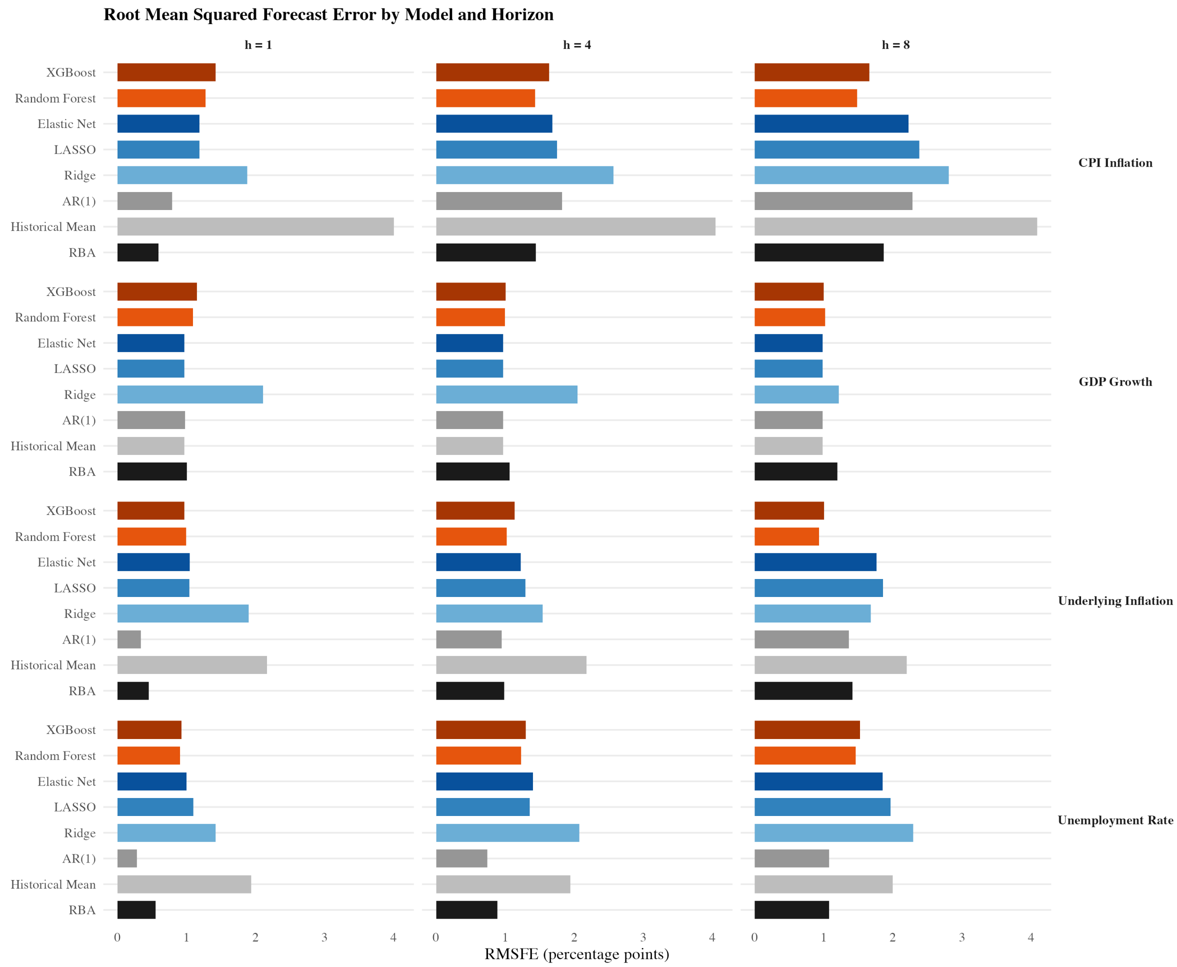

Figure 4: Root mean squared forecast error by model and horizon for all four target variables.

4.3 The Horizon Pattern

The results reveal a systematic pattern across the inflation targets. At short horizons (one to two quarters), the RBA holds a clear advantage—no machine learning model beats the RBA for CPI inflation, and only the AR(1) beats it for underlying inflation. At longer horizons (four to eight quarters), the pattern reverses: tree-based methods outperform the RBA for both CPI and underlying inflation.

We propose two complementary explanations. First, at short horizons, the RBA benefits from information that is not in the public data panel. The Bank's liaison programme, internal models, and access to high-frequency administrative data provide a short-run information advantage that no publicly available dataset can replicate. This advantage is most pronounced for headline CPI, where knowledge of administered prices, seasonal factors, and specific expenditure components matters.

Second, at longer horizons, a different dynamic may be at work. The RBA is an endogenous actor in the economy: it sets interest rates with the explicit aim of returning inflation to the 2–3 per cent target band. Consequently, the Bank's longer-horizon inflation forecasts may, at least partially, reflect the intended outcome of its own policy actions rather than a pure statistical prediction of where inflation will be. When structural forces push inflation persistently away from target—as occurred during the post-pandemic inflationary episode—the RBA's target-anchored forecasts can be systematically wrong for extended periods. Machine learning models, which have no knowledge of the RBA's policy intentions, may be better placed to identify persistent deviations from the inflation target.

For GDP growth, the pattern is different. The RBA's forecasts add relatively little beyond simple benchmarks at any horizon, echoing the finding of Tulip & Wallace (2012) over a decade earlier. For the unemployment rate, the AR(1) is more accurate than the RBA at short horizons, with the advantage converging toward zero at longer horizons.

4.4 Statistical Significance

The Diebold–Mariano test results (reported in full in Appendix C) confirm statistical significance in a minority of cases. The strongest results are for the AR(1) against the RBA:

the AR(1) is significantly more accurate for underlying inflation at $h = 0$–2 ($p < 0.05$) and for the unemployment rate at $h = 1$ ($p = 0.027$). For GDP growth at the nowcast, multiple machine learning models are significantly more accurate than the RBA ($p < 0.001$).

At longer horizons, where some of the largest relative RMSFE improvements are observed, the DM tests generally fail to reject equal accuracy. For example, the Random Forest achieves a 34 per cent improvement over the RBA for underlying inflation at $h = 8$, but the DM test has only 74 observations and uses 7 autocovariance lags in the HAC estimator, substantially reducing effective degrees of freedom. This power limitation is a fundamental challenge in macroeconomic forecast evaluation: quarterly data over three decades yields at most 130 evaluation observations at the shortest horizons, falling to 72–77 at the longest.

5 Robustness

5.1 Pre-COVID vs Post-COVID

The most important robustness check is whether the results are driven by the volatility of the COVID-19 period. Figure 5 reports RMSFE separately for the pre-COVID period (before 2020 Q1) and the post-COVID period (2020 Q1 onward).

In the pre-COVID period, differences between the RBA and machine learning models are small for all targets. The RBA's forecasts perform well in a relatively stable macroeconomic environment, and the additional complexity of machine learning adds little.

In the post-COVID period, forecast errors increase for all methods and all targets. The RBA's errors are particularly large for inflation, consistent with the supply-side shocks and behavioural changes documented in Reserve Bank of Australia (2022). Machine learning models also deteriorate, but the LASSO and tree-based methods exhibit somewhat smaller increases for inflation, suggesting that data-driven variable selection may provide partial resilience to structural breaks.

The post-COVID sample is short, and drawing conclusions about model superiority from a single extraordinary episode would not be warranted. What can be stated is that no method performed well during the pandemic, and that the relative rankings of methods are sensitive to the inclusion of this period.

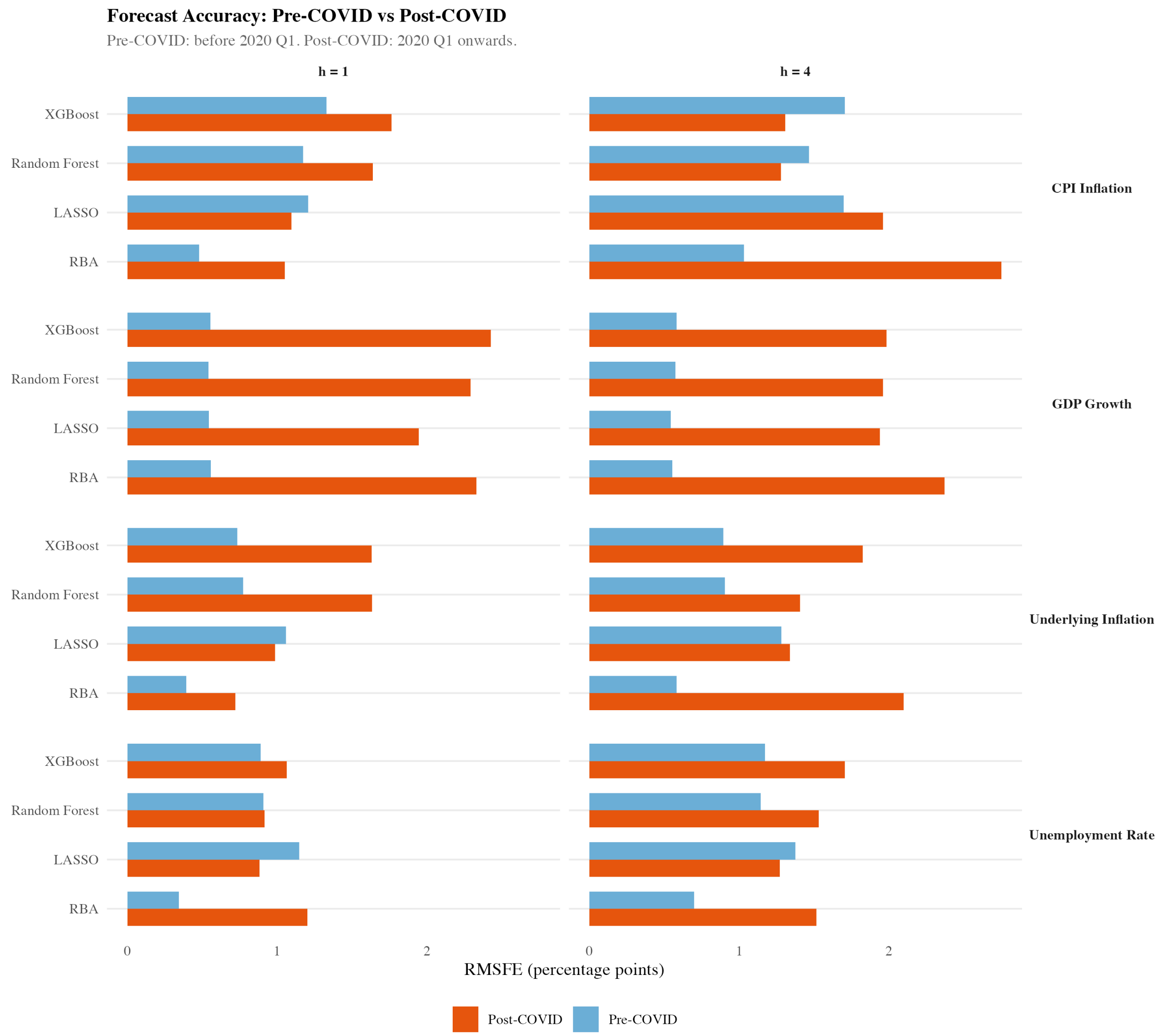

Figure 5: Forecast accuracy comparison between pre-COVID and post-COVID periods for selected models.

5.2 Sensitivity to Model Specification

The main results are robust to reasonable variations in the estimation procedure. Increasing the minimum training sample from 40 to 60 observations does not materially affect the relative rankings. Using BIC-based selection along the `glmnet` regularisation path produces results comparable to cross-validation-based selection. For the Random Forest, increasing the number of trees from 50 to 500 has negligible effect on forecast accuracy, consistent with the convergence properties of bagged ensemble methods (Breiman, 2001). Allowing deeper trees in XGBoost (maximum depth 5 rather than 3) increases RMSFE at longer horizons, consistent with overfitting.

6 Discussion

6.1 Interpretation of Results

The results indicate that machine learning methods can match or outperform the RBA's published forecasts, but that the magnitude and direction of the improvement depend on the variable and the forecast horizon. The RBA's advantage at short horizons for CPI inflation suggests that the Bank's non-public information—including its liaison programme, internal high-frequency indicators, and real-time intelligence—provides a genuine forecasting edge that cannot be replicated from published data alone. At longer horizons, where this information advantage dissipates, the disciplined variable selection of tree-based methods provides a measurable improvement.

The strong performance of the AR(1) for underlying inflation and the unemployment rate deserves particular comment. These are highly persistent series: underlying inflation (by construction) strips out volatile items, and the unemployment rate moves slowly relative to its mean. A simple autoregressive model captures this persistence effectively, while the RBA's forecasts appear to over-adjust relative to the persistence in the data, particularly at short horizons.

The GDP growth results reinforce a well-established finding in the macroeconomic forecasting literature: GDP growth is approximately unpredictable at horizons beyond the current quarter (Stock & Watson, 2002). The convergence of the LASSO and Elastic Net to the historical mean at $h \geq 1$—selecting zero predictors from the 487-variable panel—provides direct evidence that no publicly available macroeconomic variable systematically helps predict Australian GDP growth even one quarter ahead. The RBA's GDP forecasts add some information at the nowcast but perform no better than the historical mean at longer horizons.

6.2 The Central Bank as Endogenous Actor

Central banks such as the RBA are not passive observers of the economy. The RBA sets monetary policy with the explicit objective of achieving 2–3 per cent inflation over time. This creates a fundamental asymmetry in the forecasting problem. At short horizons, the RBA has leading indicators—from its liaison programme, the yield curve, and real-time administrative data—that allow it to predict economic conditions one to two quarters ahead with reasonable accuracy. At longer horizons, however, the RBA's own policy actions shape the outcomes being forecast. The Bank's two-year-ahead inflation forecast is, in part, a statement about the intended outcome of its own policy stance.

This endogeneity has implications for forecast evaluation. If the RBA's long-horizon forecasts tend toward the inflation target, they will appear accurate during periods of macroeconomic stability when inflation remains near target. But when persistent shocks drive inflation away from the target band—as in the post-pandemic period—these target-anchored forecasts can be wrong for an extended period. Machine learning models, which form expectations purely from observed data patterns without knowledge of the policy target, may be better positioned to detect persistent deviations.

6.3 Limitations

Several limitations of the analysis should be noted.

First, the comparison is not entirely symmetric. Machine learning models are trained and evaluated against the latest available release of the data, while the RBA's forecasts were made against data that have subsequently been revised. This introduces a look-ahead element whose direction is ambiguous.

Second, the exercise abstracts from the real-time challenge of model selection. In practice, a forecaster would need to decide which machine learning model to use in real time—a problem that introduces its own sources of error. Our results compare each model's full-sample performance, whereas a real-time implementation would require an additional selection step.

Third, the evaluation sample at long horizons is small (72–77 observations at $h = 8$), which limits the power of statistical tests. Some of the largest RMSFE improvements cannot be confirmed as statistically significant.

Fourth, central bank forecasts serve purposes beyond point prediction, including communicating the Bank's economic outlook and providing a framework for policy deliberation. An algorithm cannot fulfil these functions.

7 Conclusion

This paper has compared the RBA's published macroeconomic forecasts against seven machine learning and benchmark models using a panel of 487 time series and an expanding-window pseudo-real-time evaluation design over 1993–2025.

The main findings are as follows. For headline CPI inflation, the RBA outperforms all machine learning models at short horizons (one to two quarters) but is outperformed by tree-based methods at longer horizons (four to eight quarters). For underlying inflation, a simple AR(1) model outperforms the RBA at all horizons, and Random Forest achieves a 34 per cent improvement at the two-year horizon. For GDP growth, simple benchmarks match or outperform the RBA at all horizons beyond the nowcast. For the unemployment rate, the AR(1) provides large improvements at short horizons, with the advantage converging toward zero at the two-year horizon.

These results support the integration of machine learning methods within central bank forecasting frameworks, not as a replacement for expert judgement, but as a complement that can identify when model-based projections diverge from what the data suggest. The finding that the RBA's informational advantage is concentrated at short horizons, while data-driven methods gain at longer horizons, suggests that a hybrid approach—combining the Bank's short-run intelligence with algorithmic longer-run projections—could improve overall forecast accuracy.

All code and data required for replication are publicly available at github.com/andybridger/RBAvsMachine.

Appendix A: Variable Categories

Table 2: Categories of variables in the Australian macroeconomic dataset.

Category	Number of Series
GDP and national accounts	147
Prices and inflation	120
Interest rates	41
Household and business balance sheets	36
Exchange rates	28
Employment	20
Money and credit	18
Housing	15
Wages	14
Commodities	13
International	13
Industrial production and sales	8
Unemployment rate	7
Inventories	3
Equity indices	2
Expectations and surveys	1
Other	1

Appendix B: Complete Data Series

The following table lists all 487 time series in the panel, following the format of McCracken & Ng (2016). The transformation code (Tcode) column indicates the stationarity transformation applied: 1 = level, 2 = first difference, 5 = log first difference, 6 = log second difference. See Table 1 for the complete mapping.

GDP and national accounts (147 series)

ID	Series ID	Description	Source	Freq	Tcode
2	A124830484V	Public ; Final demand: Chain volume measures ; (SA)	ABS	Q	5
3	A124830485W	Private ; Final demand: Chain volume measures ; (SA)	ABS	Q	5
4	A124830486X	Public ; Final demand: Current prices ; (SA)	ABS	Q	5
5	A124830487A	Private ; Final demand: Current prices ; (SA)	ABS	Q	5
44	A2298712J	General government ; Gross operating surplus ; (SA)	ABS	Q	6
45	A2302586T	Gross domestic product - Expenditure based: Chain volume measures ; (SA)	ABS	Q	5
46	A2302587V	Gross domestic product - Income based: Chain volume measures ; (SA)	ABS	Q	5
47	A2302588W	Gross domestic product - Production based: Chain volume measures ; (SA)	ABS	Q	5
48	A2302589X	Non-farm ; Gross domestic product: Chain volume measures ; (SA)	ABS	Q	5
49	A2302590J	Non-farm ; Gross domestic product: Current prices ; (SA)	ABS	Q	5
51	A2302592L	Farm ; Gross domestic product: Chain volume measures ; (SA)	ABS	Q	1
52	A2302593R	Farm ; Gross domestic product: Current prices ; (SA)	ABS	Q	5
54	A2302597X	Private non-farm inventory levels: Chain volume measures ; (SA)	ABS	Q	5
59	A2302602F	Imports of merchandise goods: Current prices ; (SA)	ABS	Q	5
60	A2302603J	Imports to domestic sales: Ratio ; (SA)	ABS	Q	5
61	A2302604K	Wages share of total factor income: Ratio ; (SA)	ABS	Q	5
62	A2302605L	Profits share of total factor income: Ratio ; (SA)	ABS	Q	5
66	A2303246R	Food: Chain volume measures ; (SA)	ABS	Q	5
67	A2303248V	Cigarettes and tobacco: Chain volume measures ; (SA)	ABS	Q	5
68	A2303250F	Alcoholic beverages: Chain volume measures ; (SA)	ABS	Q	5

(continued)

ID	Series ID	Description	Source	Freq	Tcode
69	A2303252K	Clothing and footwear: Chain volume measures ; (SA)	ABS	Q	5
70	A2303254R	Rent and other dwelling services: Chain volume measures ; (SA)	ABS	Q	5
71	A2303256V	Electricity, gas and other fuel: Chain volume measures ; (SA)	ABS	Q	5
73	A2303260K	Health: Chain volume measures ; (SA)	ABS	Q	5
74	A2303262R	Purchase of vehicles: Chain volume measures ; (SA)	ABS	Q	5
75	A2303264V	Operation of vehicles: Chain volume measures ; (SA)	ABS	Q	5
76	A2303266X	Transport services: Chain volume measures ; (SA)	ABS	Q	1
77	A2303268C	Communications: Chain volume measures ; (SA)	ABS	Q	5
78	A2303270R	Recreation and culture: Chain volume measures ; (SA)	ABS	Q	5
79	A2303272V	Education services: Chain volume measures ; (SA)	ABS	Q	5
80	A2303274X	Hotels, cafes and restaurants: Chain volume measures ; (SA)	ABS	Q	5
81	A2303276C	Insurance and other financial services: Chain volume measures ; (SA)	ABS	Q	5
82	A2303278J	Other goods and services: Chain volume measures ; (SA)	ABS	Q	5
83	A2303280V	FINAL CONSUMPTION EXPENDITURE: Chain volume measures ; (SA)	ABS	Q	5
87	A2303363A	Public non-financial corporations ; Gross operating surplus ; (SA)	ABS	Q	5
88	A2303365F	Non-financial corporations ; Gross operating surplus ; (SA)	ABS	Q	5
89	A2303367K	Financial corporations ; Gross operating surplus ; (SA)	ABS	Q	5
90	A2303369R	Total corporations ; Gross operating surplus ; (SA)	ABS	Q	5
91	A2303373F	Dwellings owned by persons ; Gross operating surplus ; (SA)	ABS	Q	6
92	A2303375K	All sectors ; Gross operating surplus ; (SA)	ABS	Q	5
93	A2303377R	Gross mixed income ; (SA)	ABS	Q	5
94	A2303379V	Total factor income ; (SA)	ABS	Q	5
95	A2303381F	Taxes less subsidies on production and imports ; (SA)	ABS	Q	2
96	A2303383K	Statistical discrepancy (I) ; (SA)	ABS	Q	1
122	A2303730T	GROSS DOMESTIC PRODUCT ; (SA)	ABS	Q	6
123	A2303827L	General government - National ; Final consumption expenditure - Defence ; (ORIG)	ABS	Q	5
124	A2303828R	General government - National ; Final consumption expenditure - Non-defence ; (ORIG)	ABS	Q	5
125	A2303829T	General government - National ; Final consumption expenditure ; (ORIG)	ABS	Q	5

(continued)

ID	Series ID	Description	Source	Freq	Tcode
126	A2303830A	General government - State and local ; Final consumption expenditure ; (ORIG)	ABS	Q	6
127	A2303831C	General government ; Final consumption expenditure ; (ORIG)	ABS	Q	6
129	A2303833J	All sectors ; Final consumption expenditure ; (ORIG)	ABS	Q	6
130	A2303834K	Private ; Gross fixed capital formation - Machinery and equipment - Total ; (ORIG)	ABS	Q	5
131	A2303835L	Private ; Gross fixed capital formation - Non-dwelling construction - New building ; (ORIG)	ABS	Q	5
132	A2303836R	Private ; Gross fixed capital formation - Non-dwelling construction - New engineering construction ; (ORIG)	ABS	Q	5
133	A2303837T	Private ; Gross fixed capital formation - Non-dwelling construction - Total ; (ORIG)	ABS	Q	6
134	A2303839W	Private ; Gross fixed capital formation - Intellectual property products - Computer software ; (ORIG)	ABS	Q	5
135	A2303840F	Private ; Gross fixed capital formation - Intellectual property products - Mineral and petroleum exploration ; (ORIG)	ABS	Q	5
136	A2303841J	Private ; Gross fixed capital formation - Intellectual property products - Artistic originals ; (ORIG)	ABS	Q	6
137	A2303843L	Private ; Gross fixed capital formation - Total private business investment ; (ORIG)	ABS	Q	5
138	A2303844R	Private ; Gross fixed capital formation - Dwellings - New and Used ; (ORIG)	ABS	Q	5
139	A2303845T	Private ; Gross fixed capital formation - Dwellings - Alterations and additions ; (ORIG)	ABS	Q	5
140	A2303846V	Private ; Gross fixed capital formation - Dwellings - Total ; (ORIG)	ABS	Q	5
141	A2303847W	Private ; Gross fixed capital formation - Ownership transfer costs ; (ORIG)	ABS	Q	5
142	A2303848X	Private ; Gross fixed capital formation ; (ORIG)	ABS	Q	5
143	A2303849A	Public corporations - Commonwealth ; Gross fixed capital formation ; (ORIG)	ABS	Q	5
144	A2303850K	Public corporations - State and local ; Gross fixed capital formation ; (ORIG)	ABS	Q	5

(continued)

ID	Series ID	Description	Source	Freq	Tcode
145	A2303851L	Public corporations ; Gross fixed capital formation ; (ORIG)	ABS	Q	5
146	A2303852R	General government - National ; Gross fixed capital formation - Defence ; (ORIG)	ABS	Q	5
147	A2303853T	General government - National ; Gross fixed capital formation - Non-defence ; (ORIG)	ABS	Q	5
148	A2303854V	General government - National ; Gross fixed capital formation ; (ORIG)	ABS	Q	5
149	A2303855W	General government - State and local ; Gross fixed capital formation ; (ORIG)	ABS	Q	5
150	A2303856X	General government ; Gross fixed capital formation ; (ORIG)	ABS	Q	5
151	A2303857A	Public ; Gross fixed capital formation ; (ORIG)	ABS	Q	5
152	A2303858C	All sectors ; Gross fixed capital formation ; (ORIG)	ABS	Q	5
153	A2303859F	Domestic final demand ; (ORIG)	ABS	Q	6
154	A2303860R	Exports of goods and services ; (ORIG)	ABS	Q	5
155	A2303861T	Imports of goods and services ; (ORIG)	ABS	Q	5
156	A2303862V	GROSS DOMESTIC PRODUCT ; (ORIG)	ABS	Q	5
166	A2304076C	General government - National ; Final consumption expenditure - Defence ; (SA) (GDP, Quarterly)	ABS	Q	5
167	A2304077F	General government - National ; Final consumption expenditure - Non-defence ; (SA) (GDP, Quarterly)	ABS	Q	5
168	A2304078J	General government - National ; Final consumption expenditure ; (SA) (GDP, Quarterly)	ABS	Q	5
169	A2304079K	General government - State and local ; Final consumption expenditure ; (SA) (GDP, Quarterly)	ABS	Q	5
170	A2304080V	General government ; Final consumption expenditure ; (SA) (GDP, Quarterly)	ABS	Q	5
172	A2304082X	All sectors ; Final consumption expenditure ; (SA) (GDP, Quarterly)	ABS	Q	5
173	A2304083A	Private ; Gross fixed capital formation - Machinery and equipment - New ; (SA)	ABS	Q	5
174	A2304084C	Private ; Gross fixed capital formation - Machinery and equipment - Net purchase of second hand assets ; (SA)	ABS	Q	2

(continued)

ID	Series ID	Description	Source	Freq	Tcode
175	A2304085F	Private ; Gross fixed capital formation - Machinery and equipment - Total ; (SA) (GDP, Quarterly)	ABS	Q	5
176	A2304086J	Private ; Gross fixed capital formation - Non-dwelling construction - New building ; (SA) (GDP, Quarterly)	ABS	Q	5
177	A2304087K	Private ; Gross fixed capital formation - Non-dwelling construction - New engineering construction ; (SA) (GDP, Quarterly)	ABS	Q	5
178	A2304088L	Private ; Gross fixed capital formation - Non-dwelling construction - Net purchase of second hand assets ; (SA)	ABS	Q	2
179	A2304089R	Private ; Gross fixed capital formation - Non-dwelling construction - Total ; (SA) (GDP, Quarterly)	ABS	Q	5
180	A2304091A	Private ; Gross fixed capital formation - Intellectual property products - Computer software ; (SA) (GDP, Quarterly)	ABS	Q	5
181	A2304092C	Private ; Gross fixed capital formation - Intellectual property products - Mineral and petroleum exploration ; (SA) (GDP, Quarterly)	ABS	Q	5
182	A2304093F	Private ; Gross fixed capital formation - Intellectual property products - Artistic originals ; (SA) (GDP, Quarterly)	ABS	Q	5
183	A2304095K	Private ; Gross fixed capital formation - Total private business investment ; (SA) (GDP, Quarterly)	ABS	Q	5
184	A2304096L	Private ; Gross fixed capital formation - Dwellings - New and Used ; (SA) (GDP, Quarterly)	ABS	Q	5
185	A2304097R	Private ; Gross fixed capital formation - Dwellings - Alterations and additions ; (SA) (GDP, Quarterly)	ABS	Q	1
186	A2304098T	Private ; Gross fixed capital formation - Dwellings - Total ; (SA) (GDP, Quarterly)	ABS	Q	1
187	A2304099V	Private ; Gross fixed capital formation - Ownership transfer costs ; (SA) (GDP, Quarterly)	ABS	Q	5
188	A2304100T	Private ; Gross fixed capital formation ; (SA) (GDP, Quarterly)	ABS	Q	5
189	A2304101V	Public corporations - Commonwealth ; Gross fixed capital formation ; (SA) (GDP, Quarterly)	ABS	Q	5
190	A2304102W	Public corporations - State and local ; Gross fixed capital formation ; (SA) (GDP, Quarterly)	ABS	Q	2

(continued)

ID	Series ID	Description	Source	Freq	Tcode
191	A2304103X	Public corporations ; Gross fixed capital formation ; (SA) (GDP, Quarterly)	ABS	Q	2
192	A2304104A	General government - National ; Gross fixed capital formation - Defence ; (SA) (GDP, Quarterly)	ABS	Q	5
193	A2304105C	General government - National ; Gross fixed capital formation - Non-defence ; (SA) (GDP, Quarterly)	ABS	Q	2
194	A2304106F	General government - National ; Gross fixed capital formation ; (SA) (GDP, Quarterly)	ABS	Q	5
195	A2304107J	General government - State and local ; Gross fixed capital formation ; (SA) (GDP, Quarterly)	ABS	Q	5
196	A2304108K	General government ; Gross fixed capital formation ; (SA) (GDP, Quarterly)	ABS	Q	5
197	A2304109L	Public ; Gross fixed capital formation ; (SA) (GDP, Quarterly)	ABS	Q	5
198	A2304110W	All sectors ; Gross fixed capital formation ; (SA) (GDP, Quarterly)	ABS	Q	5
199	A2304111X	Domestic final demand ; (SA) (GDP, Quarterly)	ABS	Q	5
201	A2304113C	Gross national expenditure ; (SA) (GDP, Quarterly)	ABS	Q	5
202	A2304114F	Exports of goods and services ; (SA) (GDP, Quarterly)	ABS	Q	5
203	A2304115J	Imports of goods and services ; (SA) (GDP, Quarterly)	ABS	Q	5
204	A2304116K	Statistical discrepancy (E) ; (SA)	ABS	Q	1
206	A2304192L	GDP per hour worked: Index ; (SA)	ABS	Q	5
207	A2304196W	Gross domestic product: Index ; (SA)	ABS	Q	5
208	A2304198A	Domestic final demand: Index ; (SA)	ABS	Q	6
209	A2304200A	Terms of trade: Index ; (SA)	ABS	Q	5
210	A2304402X	Gross domestic product: Chain volume measures ; (SA)	ABS	Q	5
211	A2304404C	GDP per capita: Chain volume measures ; (SA)	ABS	Q	5
212	A2304408L	Net domestic product: Chain volume measures ; (SA)	ABS	Q	5
213	A2304410X	Real gross domestic income: Chain volume measures ; (SA)	ABS	Q	5
214	A2304412C	Real gross national income: Chain volume measures ; (SA)	ABS	Q	5
215	A2304414J	Real net national disposable income: Chain volume measures ; (SA)	ABS	Q	5
216	A2304416L	Real net national disposable income per capita: Chain volume measures ; (SA)	ABS	Q	5

(continued)

ID	Series ID	Description	Source	Freq	Tcode
217	A2304418T	Gross domestic product: Current prices ; (SA)	ABS	Q	5
218	A2304420C	GDP per capita: Current prices ; (SA)	ABS	Q	5
219	A2304422J	Gross national income: Current prices ; (SA)	ABS	Q	5
220	A2304424L	Net saving: Current prices ; (SA)	ABS	Q	2
222	A2305146T	Gross fixed capital formation - New private business investment: Current prices ; (SA)	ABS	Q	5
223	A2305148W	Gross fixed capital formation - New private business investment: Chain volume measures ; (SA)	ABS	Q	5
225	A2323372A	Private non-financial corporations ; Gross operating surplus ; (SA)	ABS	Q	5
277	A2716197L	Private ; Gross fixed capital formation - Cultivated biological resources ; (SA) (GDP, Quarterly)	ABS	Q	1
278	A2716198R	Private ; Gross fixed capital formation - Intellectual property products ; (SA) (GDP, Quarterly)	ABS	Q	5
279	A2716199T	Private ; Gross fixed capital formation - Intellectual property products - Research and development ; (SA) (GDP, Quarterly)	ABS	Q	5
280	A2716238W	Private ; Gross fixed capital formation - Cultivated biological resources ; (ORIG)	ABS	Q	5
281	A2716239X	Private ; Gross fixed capital formation - Intellectual property products ; (ORIG)	ABS	Q	5
282	A2716240J	Private ; Gross fixed capital formation - Intellectual property products - Research and development ; (ORIG)	ABS	Q	5
286	A3531609C	Gross Operating Profits ; Total (State) ; Total (Industry) ; Current Price ; CORP ; (SA)	ABS	Q	5
287	A3531621V	Gross Operating Profits ; Total (State) ; Mining ; Current Price ; CORP ; (SA)	ABS	Q	5
293	A3606056V	Gross value added market sector: Chain volume measures ; (SA)	ABS	Q	5
294	A3606058X	Gross value added per hour worked market sector: Index ; (SA)	ABS	Q	5
307	A85125298V	Gross fixed capital formation - Mining private business investment: Current prices ; (SA)	ABS	Q	6
308	A85125299W	Gross fixed capital formation - Non-mining private business investment: Current prices ; (SA)	ABS	Q	6
313	A85222698X	Gross fixed capital formation - Mining private business investment: Chain volume measures ; (SA)	ABS	Q	6

(continued)

ID	Series ID	Description	Source	Freq	Tcode
314	A85222699A	Gross fixed capital formation - Non-mining private business investment: Chain volume measures ; (SA)	ABS	Q	5

Prices and inflation (120 series)

ID	Series ID	Description	Source	Freq	Tcode
6	A128473239F	Index Numbers ; All groups CPI excluding 'volatile items' ; Australia ; (ORIG) (Prices, Monthly)	ABS	M	6
7	A128473315W	Index Numbers ; Health ; Australia ; (ORIG) (Prices, Monthly)	ABS	M	5
8	A128473361K	Index Numbers ; Education ; Australia ; (ORIG) (Prices, Monthly)	ABS	M	5
9	A128473449C	Index Numbers ; Rents ; Australia ; (ORIG)	ABS	M	6
10	A128473623X	Index Numbers ; Non-alcoholic beverages ; Australia ; (ORIG)	ABS	M	5
11	A128474303F	Index Numbers ; Electricity ; Australia ; (ORIG)	ABS	M	5
12	A128475063X	Index Numbers ; Communication ; Australia ; (ORIG) (Prices, Monthly)	ABS	M	5
13	A128475329W	Index Numbers ; Food products n.e.c. ; Australia ; (ORIG)	ABS	M	6
14	A128475405L	Index Numbers ; New dwelling purchase by owner-occupiers ; Australia ; (ORIG)	ABS	M	6
15	A128475583C	Index Numbers ; Clothing and footwear ; Australia ; (ORIG) (Prices, Monthly)	ABS	M	5
16	A128475629X	Index Numbers ; Bread and cereal products ; Australia ; (ORIG)	ABS	M	6
17	A128475797K	Index Numbers ; Holiday travel and accommodation ; Australia ; (ORIG)	ABS	M	5
18	A128476505R	Index Numbers ; Tradables ; Australia ; (ORIG) (Prices, Monthly)	ABS	M	5
19	A128476849V	Index Numbers ; Gas and other household fuels ; Australia ; (ORIG)	ABS	M	5
20	A128477043A	Index Numbers ; Insurance and financial services ; Australia ; (ORIG) (Prices, Monthly)	ABS	M	6
21	A128477399R	Index Numbers ; Alcohol and tobacco ; Australia ; (ORIG) (Prices, Monthly)	ABS	M	5
22	A128477405X	Index Numbers ; Dairy and related products ; Australia ; (ORIG)	ABS	M	6

(continued)

ID	Series ID	Description	Source	Freq	Tcode
23	A128478317T	Index Numbers ; All groups CPI ; Australia ; (ORIG) (Prices, Monthly)	ABS	M	6
24	A128478473V	Index Numbers ; Recreation and culture ; Australia ; (ORIG) (Prices, Monthly)	ABS	M	5
25	A128478847A	Index Numbers ; Garments ; Australia ; (ORIG)	ABS	M	5
26	A128479215W	Index Numbers ; Meat and seafoods ; Australia ; (ORIG)	ABS	M	5
27	A128479281V	Index Numbers ; Alcoholic beverages ; Australia ; (ORIG)	ABS	M	5
28	A128480133R	Index Numbers ; Non-tradables ; Australia ; (ORIG) (Prices, Monthly)	ABS	M	5
29	A128480215V	Index Numbers ; Fruit and vegetables ; Australia ; (ORIG)	ABS	M	1
30	A128480733V	Index Numbers ; Food and non-alcoholic beverages ; Australia ; (ORIG) (Prices, Monthly)	ABS	M	5
31	A128481587A	Index Numbers ; All groups CPI, seasonally adjusted ; Australia ; (SA) (Prices, Monthly)	ABS	M	6
32	A128481639T	Index Numbers ; All groups, services component ; Australia ; (ORIG) (Prices, Monthly)	ABS	M	5
33	A128482435J	Index Numbers ; Transport ; Australia ; (ORIG) (Prices, Monthly)	ABS	M	5
34	A128483461F	Index Numbers ; All groups, goods component ; Australia ; (ORIG) (Prices, Monthly)	ABS	M	6
35	A128484257R	Index Numbers ; Tobacco ; Australia ; (ORIG)	ABS	M	5
36	A128484685K	Index Numbers ; Automotive fuel ; Australia ; (ORIG)	ABS	M	5
37	A128485865L	Index Numbers ; Housing ; Australia ; (ORIG) (Prices, Monthly)	ABS	M	6
38	A128485881L	Index Numbers ; Furnishings, household equipment and services ; Australia ; (ORIG) (Prices, Monthly)	ABS	M	5
41	A130184499R	Index Numbers ; All groups CPI excluding 'volatile items' and holiday travel ; Australia ; (ORIG)	ABS	M	5
42	A130184501R	Index Numbers ; All groups CPI excluding 'volatile items' and holiday travel ; Australia ; (SA)	ABS	M	5
50	A2302591K	Non-farm ; Gross domestic product: Implicit price deflators ; (SA)	ABS	Q	5
53	A2302594T	Farm ; Gross domestic product: Implicit price deflators ; (SA)	ABS	Q	5

(continued)

ID	Series ID	Description	Source	Freq	Tcode
97	A2303703K	Private ; Gross fixed capital formation - Non-dwelling construction - New engineering construction ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	5
98	A2303704L	Private ; Gross fixed capital formation - Non-dwelling construction - Total ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	5
99	A2303706T	Private ; Gross fixed capital formation - Intellectual property products - Computer software ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	5
100	A2303707V	Private ; Gross fixed capital formation - Intellectual property products - Mineral and petroleum exploration ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	5
101	A2303708W	Private ; Gross fixed capital formation - Intellectual property products - Artistic originals ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	5
102	A2303710J	Private ; Gross fixed capital formation - Total private business investment ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	5
103	A2303711K	Private ; Gross fixed capital formation - Dwellings - New and Used ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	5
104	A2303712L	Private ; Gross fixed capital formation - Dwellings - Alterations and additions ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	5
105	A2303713R	Private ; Gross fixed capital formation - Dwellings - Total ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	5
106	A2303714T	Private ; Gross fixed capital formation - Ownership transfer costs ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	5
107	A2303715V	Private ; Gross fixed capital formation ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	5
108	A2303716W	Public corporations - Commonwealth ; Gross fixed capital formation ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	5
109	A2303717X	Public corporations - State and local ; Gross fixed capital formation ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	5
110	A2303718A	Public corporations ; Gross fixed capital formation ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	2

(continued)

ID	Series ID	Description	Source	Freq	Tcode
111	A2303719C	General government - National ; Gross fixed capital formation - Defence ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	5
112	A2303720L	General government - National ; Gross fixed capital formation - Non-defence ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	5
113	A2303721R	General government - National ; Gross fixed capital formation ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	5
114	A2303722T	General government - State and local ; Gross fixed capital formation ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	5
115	A2303723V	General government ; Gross fixed capital formation ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	5
116	A2303724W	Public ; Gross fixed capital formation ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	5
117	A2303725X	All sectors ; Gross fixed capital formation ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	5
118	A2303726A	Domestic final demand ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	5
119	A2303727C	Gross national expenditure ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	6
120	A2303728F	Exports of goods and services ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	5
121	A2303729J	Imports of goods and services ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	5
157	A2303935W	General government - National ; Final consumption expenditure - Defence ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	5
158	A2303936X	General government - National ; Final consumption expenditure - Non-defence ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	5
159	A2303937A	General government - National ; Final consumption expenditure ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	5
160	A2303938C	General government - State and local ; Final consumption expenditure ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	5
161	A2303939F	General government ; Final consumption expenditure ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	6

(continued)

ID	Series ID	Description	Source	Freq	Tcode
162	A2303940R	Households ; Final consumption expenditure ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	6
163	A2303941T	All sectors ; Final consumption expenditure ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	6
164	A2303942V	Private ; Gross fixed capital formation - Machinery and equipment - Total ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	5
165	A2303943W	Private ; Gross fixed capital formation - Non-dwelling construction - New building ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	5
224	A2314865F	Index Numbers ; Final ; Total (Source) ; (ORIG)	ABS	Q	5
227	A2325846C	Index Numbers ; All groups CPI ; Australia ; (ORIG) (Prices, Quarterly)	ABS	Q	5
228	A2325891R	Index Numbers ; Food and non-alcoholic beverages ; Australia ; (ORIG) (Prices, Quarterly)	ABS	Q	5
229	A2325936J	Index Numbers ; Clothing and footwear ; Australia ; (ORIG) (Prices, Quarterly)	ABS	Q	6
230	A2325981V	Index Numbers ; Housing ; Australia ; (ORIG) (Prices, Quarterly)	ABS	Q	5
231	A2326026R	Index Numbers ; Furnishings, household equipment and services ; Australia ; (ORIG) (Prices, Quarterly)	ABS	Q	5
232	A2326071A	Index Numbers ; Transport ; Australia ; (ORIG) (Prices, Quarterly)	ABS	Q	5
233	A2326116V	Index Numbers ; Alcohol and tobacco ; Australia ; (ORIG) (Prices, Quarterly)	ABS	Q	5
234	A2330526L	Index Numbers ; Tradables ; Australia ; (ORIG) (Prices, Quarterly)	ABS	Q	6
235	A2330571X	Index Numbers ; Non-tradables ; Australia ; (ORIG) (Prices, Quarterly)	ABS	Q	5
236	A2330616T	Index Numbers ; All groups, goods component ; Australia ; (ORIG) (Prices, Quarterly)	ABS	Q	5
237	A2330661C	Index Numbers ; Market goods and services excluding 'volatile items' - Goods ; Australia ; (ORIG)	ABS	Q	5
238	A2330706W	Index Numbers ; All groups, services component ; Australia ; (ORIG) (Prices, Quarterly)	ABS	Q	5

(continued)

ID	Series ID	Description	Source	Freq	Tcode
239	A2330751J	Index Numbers ; Market goods and services excluding 'volatile items' - Services ; Australia ; (ORIG)	ABS	Q	5
240	A2330796L	Index Numbers ; Market goods and services excluding 'volatile items' - Total ; Australia ; (ORIG)	ABS	Q	5
241	A2330841L	Index Numbers ; All groups CPI excluding 'volatile items' ; Australia ; (ORIG) (Prices, Quarterly)	ABS	Q	6
242	A2331111C	Index Numbers ; Health ; Australia ; (ORIG) (Prices, Quarterly)	ABS	Q	5
243	A2331201J	Index Numbers ; Communication ; Australia ; (ORIG) (Prices, Quarterly)	ABS	Q	5
244	A2331246L	Index Numbers ; Recreation and culture ; Australia ; (ORIG) (Prices, Quarterly)	ABS	Q	5
245	A2331426W	Index Numbers ; Education ; Australia ; (ORIG) (Prices, Quarterly)	ABS	Q	6
246	A2332101T	Index Numbers ; All groups CPI excluding Food and non-alcoholic beverages ; Australia ; (ORIG)	ABS	Q	5
247	A2332146W	Index Numbers ; All groups CPI excluding Alcohol and tobacco ; Australia ; (ORIG)	ABS	Q	6
248	A2332191J	Index Numbers ; All groups CPI excluding Clothing and footwear ; Australia ; (ORIG)	ABS	Q	5
249	A2332236A	Index Numbers ; All groups CPI excluding Housing ; Australia ; (ORIG)	ABS	Q	6
250	A2332281L	Index Numbers ; All groups CPI excluding Furnishings, household equipment and services ; Australia ; (ORIG)	ABS	Q	5
251	A2332326F	Index Numbers ; All groups CPI excluding Health ; Australia ; (ORIG)	ABS	Q	5
252	A2332371T	Index Numbers ; All groups CPI excluding Transport ; Australia ; (ORIG)	ABS	Q	5
253	A2332416K	Index Numbers ; All groups CPI excluding Communication ; Australia ; (ORIG)	ABS	Q	5
254	A2332461W	Index Numbers ; All groups CPI excluding Recreation and culture ; Australia ; (ORIG)	ABS	Q	5
255	A2332506R	Index Numbers ; All groups CPI excluding Education ; Australia ; (ORIG)	ABS	Q	5

(continued)

ID	Series ID	Description	Source	Freq	Tcode
256	A2332551A	Index Numbers ; All groups CPI excluding Medical and hospital services ; Australia ; (ORIG)	ABS	Q	6
257	A2332596F	Index Numbers ; Insurance and financial services ; Australia ; (ORIG) (Prices, Quarterly)	ABS	Q	5
258	A2332641F	Index Numbers ; All groups CPI excluding Insurance and financial services ; Australia ; (ORIG)	ABS	Q	5
259	A2332686K	Index Numbers ; All groups CPI excluding Housing and Insurance and financial services ; Australia ; (ORIG)	ABS	Q	5
274	A2716115W	Private ; Gross fixed capital formation - Cultivated biological resources ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	5
275	A2716116X	Private ; Gross fixed capital formation - Intellectual property products ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	6
276	A2716117A	Private ; Gross fixed capital formation - Intellectual property products - Research and development ; (SA) (GDP implicit price deflators, Quarterly)	ABS	Q	5
288	A3604378X	Index Numbers ; All groups CPI excluding food and energy ; Australia ; (ORIG)	ABS	Q	5
289	A3604398J	Index Numbers ; All groups CPI including deposit and loan facilities (indirect charges) ; Australia ; (ORIG)	ABS	Q	5
290	A3604503X	Index Numbers ; Weighted Median ; Australia ; (SA)	ABS	Q	6
291	A3604506F	Index Numbers ; All groups CPI, seasonally adjusted ; Australia ; (SA) (Prices, Quarterly)	ABS	Q	6
292	A3604509L	Index Numbers ; Trimmed Mean ; Australia ; (SA)	ABS	Q	6
452	GBONYLD	Break-even 10-year inflation rate (ORIG)	RBA	Q	6
453	GBUSEXP	Business inflation expectations <U+2013> 3-months ahead (ORIG)	RBA	Q	1
455	GCONEXP	Consumer inflation expectations <U+2013> 1-year ahead (ORIG)	RBA	Q	1
464	GMAREXPY	Market economists' inflation expectations <U+2013> 1-year ahead (ORIG)	RBA	Q	1
465	GMAREXPYY	Market economists' inflation expectations <U+2013> 2-year ahead (ORIG)	RBA	Q	5
473	GUNIEXPY	Union officials' inflation expectations <U+2013> 1-year ahead (ORIG)	RBA	Q	1

(continued)

ID	Series ID	Description	Source	Freq	Tcode
474	GUNIEXPYY	Union officials’ inflation expectations <U+2013> 2-year ahead (ORIG)	RBA	Q	5

Interest rates (41 series)

ID	Series ID	Description	Source	Freq	Tcode
347	BSPNSPNLD	Business bonds liabilities (ORIG)	RBA	Q	5
370	FCMYGBAG10	Australian Government 10 year bond (ORIG)	RBA	M	5
371	FCMYGBAG2	Australian Government 2 year bond (ORIG)	RBA	M	5
372	FCMYGBAG3	Australian Government 3 year bond (ORIG)	RBA	M	5
373	FCMYGBAG5	Australian Government 5 year bond (ORIG)	RBA	M	5
382	FILRPLRCCL	Lending rates; Personal loans; Revolving credit; Credit cards; Low rate (ORIG)	RBA	M	5
383	FILRPLRCCS	Lending rates; Personal loans; Revolving credit; Credit cards; Standard (ORIG)	RBA	M	1
384	FILRSBVOO	Lending rates; Small business; Variable; Overdraft (ORIG)	RBA	M	5
385	FILRSBVRT	Lending rates; Small business; Variable; Term (ORIG)	RBA	M	1
386	FIRMMBAB180	6-month BABs/NCDs (ORIG)	RBA	M	5
387	FIRMMBAB30	1-month BABs/NCDs (ORIG)	RBA	M	5
388	FIRMMBAB90	3-month BABs/NCDs (ORIG)	RBA	M	5
389	FIRMMOIS1	1-month OIS (ORIG)	RBA	M	6
390	FIRMMOIS3	3-month OIS (ORIG)	RBA	M	6
391	FIRMMOIS6	6-month OIS (ORIG)	RBA	M	6
392	FNFNA10M	Non-financial corporate A-rated bonds <U+2013> Number of bonds <U+2013> 8<U+2013>12 years (ORIG)	RBA	M	5
393	FNFNA12M	Non-financial corporate A-rated bonds <U+2013> Number of bonds <U+2013> >12 years (ORIG)	RBA	M	2
394	FNFNA3M	Non-financial corporate A-rated bonds <U+2013> Number of bonds <U+2013> 1<U+2013>4 years (ORIG)	RBA	M	5
395	FNFNA5M	Non-financial corporate A-rated bonds <U+2013> Number of bonds <U+2013> 4<U+2013>6 years (ORIG)	RBA	M	5

(continued)

ID	Series ID	Description	Source	Freq	Tcode
396	FNFNA7M	Non-financial corporate A-rated bonds <U+2013> Number of bonds <U+2013> 6<U+2013>8 years (ORIG)	RBA	M	5
397	FNFNBBB10M	Non-financial corporate BBB-rated bonds <U+2013> Number of bonds <U+2013> 8<U+2013>12 years (ORIG)	RBA	M	1
398	FNFNBBB12M	Non-financial corporate BBB-rated bonds <U+2013> Number of bonds <U+2013> >12 years (ORIG)	RBA	M	2
399	FNFNBBB3M	Non-financial corporate BBB-rated bonds <U+2013> Number of bonds <U+2013> 1<U+2013>4 years (ORIG)	RBA	M	5
400	FNFNBBB5M	Non-financial corporate BBB-rated bonds <U+2013> Number of bonds <U+2013> 4<U+2013>6 years (ORIG)	RBA	M	5
401	FNFNBBB7M	Non-financial corporate BBB-rated bonds <U+2013> Number of bonds <U+2013> 6<U+2013>8 years (ORIG)	RBA	M	2
402	FNFTA10M	Non-financial corporate A-rated bonds <U+2013> Effective tenor <U+2013> 10 year target tenor (ORIG)	RBA	M	1
403	FNFTA3M	Non-financial corporate A-rated bonds <U+2013> Effective tenor <U+2013> 3 year target tenor (ORIG)	RBA	M	1
404	FNFTA5M	Non-financial corporate A-rated bonds <U+2013> Effective tenor <U+2013> 5 year target tenor (ORIG)	RBA	M	1
405	FNFTA7M	Non-financial corporate A-rated bonds <U+2013> Effective tenor <U+2013> 7 year target tenor (ORIG)	RBA	M	1
406	FNFTBBB10M	Non-financial corporate BBB-rated bonds <U+2013> Effective tenor <U+2013> 10 year target tenor (ORIG)	RBA	M	1
407	FNFTBBB3M	Non-financial corporate BBB-rated bonds <U+2013> Effective tenor <U+2013> 3 year target tenor (ORIG)	RBA	M	5
408	FNFTBBB5M	Non-financial corporate BBB-rated bonds <U+2013> Effective tenor <U+2013> 5 year target tenor (ORIG)	RBA	M	5
409	FNFTBBB7M	Non-financial corporate BBB-rated bonds <U+2013> Effective tenor <U+2013> 7 year target tenor (ORIG)	RBA	M	5
410	FNFYA10M	Non-financial corporate A-rated bonds <U+2013> Yield <U+2013> 10 year target tenor (ORIG)	RBA	M	5
411	FNFYA3M	Non-financial corporate A-rated bonds <U+2013> Yield <U+2013> 3 year target tenor (ORIG)	RBA	M	5

(continued)

ID	Series ID	Description	Source	Freq	Tcode
412	FNFYA5M	Non-financial corporate A-rated bonds <U+2013> Yield <U+2013> 5 year target tenor (ORIG)	RBA	M	5
413	FNFYA7M	Non-financial corporate A-rated bonds <U+2013> Yield <U+2013> 7 year target tenor (ORIG)	RBA	M	5
414	FNFYBBB10M	Non-financial corporate BBB-rated bonds <U+2013> Yield <U+2013> 10 year target tenor (ORIG)	RBA	M	5
415	FNFYBBB3M	Non-financial corporate BBB-rated bonds <U+2013> Yield <U+2013> 3 year target tenor (ORIG)	RBA	M	5
416	FNFYBBB5M	Non-financial corporate BBB-rated bonds <U+2013> Yield <U+2013> 5 year target tenor (ORIG)	RBA	M	5
417	FNFYBBB7M	Non-financial corporate BBB-rated bonds <U+2013> Yield <U+2013> 7 year target tenor (ORIG)	RBA	M	5

Household and business balance sheets (36 series)

ID	Series ID	Description	Source	Freq	Tcode
72	A2303258X	Furnishings and household equipment: Chain volume measures ; (SA)	ABS	Q	5
128	A2303832F	Households ; Final consumption expenditure ; (ORIG)	ABS	Q	6
171	A2304081W	Households ; Final consumption expenditure ; (SA) (GDP, Quarterly)	ABS	Q	5
226	A2323382F	Household saving ratio: Ratio ; (SA)	ABS	Q	2
309	A85125310X	Households ; Final consumption expenditure - Goods: Chain volume measures ; (SA)	ABS	Q	5
310	A85125311A	Households ; Final consumption expenditure - Services: Chain volume measures ; (SA)	ABS	Q	1
311	A85125312C	Households ; Final consumption expenditure - Essential consumption: Chain volume measures ; (SA)	ABS	Q	5
312	A85125313F	Households ; Final consumption expenditure - Discretionary consumption: Chain volume measures ; (SA)	ABS	Q	1
315	A85251203A	Households ; Final consumption expenditure - Goods: Current prices ; (SA)	ABS	Q	5

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ID	Series ID	Description	Source	Freq	Tcode
316	A85251204C	Households ; Final consumption expenditure - Services: Current prices ; (SA)	ABS	Q	5
317	A85251205F	Households ; Final consumption expenditure - Essential consumption: Current prices ; (SA)	ABS	Q	6
318	A85251206J	Households ; Final consumption expenditure - Discretionary consumption: Current prices ; (SA)	ABS	Q	5
321	BHFADIFA	Household financial assets to income (ORIG)	RBA	Q	5
322	BHFADIT	Household assets to income (ORIG)	RBA	Q	5
323	BHFDA	Household debt to assets (ORIG)	RBA	Q	5
326	BHFDDIT	Household debt to income (ORIG)	RBA	Q	5
329	BSPNSHNFD	Household dwellings (ORIG)	RBA	Q	5
330	BSPNSHNFO	Household consumer durables (ORIG)	RBA	Q	6
331	BSPNSHNFT	Household total non-financial assets (ORIG)	RBA	Q	5
332	BSPNSHUA	Household total assets (ORIG)	RBA	Q	5
335	BSPNSHUFAD	Household deposits (ORIG)	RBA	Q	5
336	BSPNSHUFAO	Household other financial assets (ORIG)	RBA	Q	5
337	BSPNSHUFAR	Household superannuation (ORIG)	RBA	Q	5
338	BSPNSHUFAS	Household equities (ORIG)	RBA	Q	6
339	BSPNSHUFAT	Household total financial assets (ORIG)	RBA	Q	5
340	BSPNSHUL	Household total liabilities (ORIG)	RBA	Q	6
341	BSPNSHUNW	Household net worth (ORIG)	RBA	Q	5
342	BSPNSPNFAD	Business bank deposits (ORIG)	RBA	Q	5
343	BSPNSPNFAF	Business foreign equities (ORIG)	RBA	Q	5
344	BSPNSPNFAO	Business other financial assets (ORIG)	RBA	Q	5
345	BSPNSPNFAT	Business total financial assets (ORIG)	RBA	Q	5
346	BSPNSPNLB	Business bills of exchange liabilities (ORIG)	RBA	Q	6
348	BSPNSPNLL	Business loans (ORIG)	RBA	Q	6
349	BSPNSPNLO	Business other liabilities (ORIG)	RBA	Q	5
350	BSPNSPNLS	Business equities liabilities (ORIG)	RBA	Q	5
351	BSPNSPNLT	Business total liabilities (ORIG)	RBA	Q	5

Exchange rates (28 series)

ID	Series ID	Description	Source	Freq	Tcode
424	FREREWI	Real export-weighted index (ORIG)	RBA	Q	5
425	FRERIWI	Real import-weighted index (ORIG)	RBA	Q	5
426	FRERTWI	Real trade-weighted index (ORIG)	RBA	Q	5
427	FXRCD	A$1=CAD (ORIG)	RBA	M	1
428	FXRCR	A$1=CNY (ORIG)	RBA	M	5
429	FXREUR	A$1=EUR (ORIG)	RBA	M	5
430	FXRHKD	A$1=HKD (ORIG)	RBA	M	5
431	FXRIR	A$1=IDR (ORIG)	RBA	M	5
432	FXRIRE	A$1=INR (ORIG)	RBA	M	5
433	FXRJY	A$1=JPY (ORIG)	RBA	M	5
434	FXRMR	A$1=MYR (ORIG)	RBA	M	5
435	FXRNTD	A$1=TWD (ORIG)	RBA	M	5
436	FXRNZD	A$1=NZD (ORIG)	RBA	M	5
437	FXRPHP	A$1=PHP (ORIG)	RBA	M	6
438	FXRPNGK	A$1=PGK (ORIG)	RBA	M	5
439	FXRSARD	A$1=ZAR (ORIG)	RBA	M	5
440	FXRSARY	A$1=SAR (ORIG)	RBA	M	5
441	FXRSD	A$1=SGD (ORIG)	RBA	M	1
442	FXRSDR	A$1=SDR (ORIG)	RBA	M	5
443	FXRSF	A$1=CHF (ORIG)	RBA	M	5
444	FXRSK	A$1=SEK (ORIG)	RBA	M	5
445	FXRSKW	A$1=KRW (ORIG)	RBA	M	5
446	FXRTB	A$1=THB (ORIG)	RBA	M	5
447	FXRTWI	Trade-weighted Index May 1970 = 100 (ORIG)	RBA	M	5
448	FXRUAED	A$1=AED (ORIG)	RBA	M	5
449	FXRUKPS	A$1=GBP (ORIG)	RBA	M	5
450	FXRUSD	A$1=USD (ORIG)	RBA	M	5
451	FXRVD	A$1=VND (ORIG)	RBA	M	5

Employment (20 series)

ID	Series ID	Description	Source	Freq	Tcode
39	A129552324A	Compensation of employees per hour: Current prices ; (SA)	ABS	Q	5

(continued)

ID	Series ID	Description	Source	Freq	Tcode
40	A129552325C	Non-farm compensation of employees per hour: Current prices ; (SA)	ABS	Q	5
63	A2302606R	Average compensation per employee: Current prices ; (SA)	ABS	Q	5
64	A2302607T	Non-farm ; Total compensation of employees: Current prices ; (SA)	ABS	Q	5
65	A2302608V	Average non-farm compensation per employee: Current prices ; (SA)	ABS	Q	5
84	A2303355A	Compensation of employees - Wages and salaries ; (SA)	ABS	Q	6
85	A2303357F	Compensation of employees - Employers' social contributions ; (SA)	ABS	Q	5
86	A2303359K	Compensation of employees ; (SA)	ABS	Q	5
205	A2304190J	Hours worked market sector: Index ; (SA)	ABS	Q	1
221	A2304428W	Hours worked: Index ; (SA)	ABS	Q	5
295	A84423041X	> Employed full-time ; Persons ; (SA)	ABS	M	5
296	A84423042A	> Employed part-time ; Persons ; (SA)	ABS	M	1
297	A84423043C	Employed total ; Persons ; (SA)	ABS	M	5
301	A84423047L	Labour force total ; Persons ; (SA)	ABS	M	5
303	A84423051C	Participation rate ; Persons ; (SA)	ABS	M	5
306	A84423054K	Employment to population ratio ; Persons ; (SA)	ABS	M	5
319	A85255725J	Underemployment rate (proportion of labour force) ; Persons ; (SA)	ABS	M	5
461	GLFOSVPS	Private sector job vacancies (SA)	RBA	Q	1
462	GLFOSVT	Job vacancies (SA)	RBA	Q	5
463	GLFOSVTLF	Vacancies to labour force ratio (SA)	RBA	Q	1

Money and credit (18 series)

ID	Series ID	Description	Source	Freq	Tcode
352	DGFACHM	Credit; Housing; Monthly growth (SA)	RBA	M	1
353	DGFACIHM	Credit; Investor housing; Monthly growth (SA)	RBA	M	1
354	DGFACOHM	Credit; Owner-occupier housing; Monthly growth (SA)	RBA	M	5
355	DGFACOPM	Credit; Other personal; Monthly growth (SA)	RBA	M	1

(continued)

ID	Series ID	Description	Source	Freq	Tcode
356	DMABMN	Broad money (ORIG)	RBA	M	5
357	DMABMS	Broad money: Seasonally adjusted (SA)	RBA	M	5
358	DMACN	Currency (ORIG)	RBA	M	5
359	DMACS	Currency: Seasonally adjusted (SA)	RBA	M	5
360	DMAM1N	M1 (ORIG)	RBA	M	5
361	DMAM1S	M1: Seasonally adjusted (SA)	RBA	M	5
362	DMAM3N	M3 (ORIG)	RBA	M	5
363	DMAM3S	M3: Seasonally adjusted (SA)	RBA	M	5
364	DMAMMB	Money base (ORIG)	RBA	M	5
365	DMAMOAFI	Offshore borrowings by AFIs (ORIG)	RBA	M	5
366	DMANBPA	Other borrowings from private sector by AFIs (ORIG)	RBA	M	5
367	DMANTD	Non-Transaction Deposits with ADIs (ORIG)	RBA	M	5
368	DMAODCD	Certificates of deposit issued by ADIs (ORIG)	RBA	M	5
369	DMATD	Transaction Deposits with ADIs (ORIG)	RBA	M	5

Housing (15 series)

ID	Series ID	Description	Source	Freq	Tcode
324	BHFDDIH	Housing debt to income (ORIG)	RBA	Q	5
325	BHFDDIO	Owner-occupier housing debt to income (SA)	RBA	Q	6
327	BHFHDHA	Housing debt to housing assets (ORIG)	RBA	Q	5
328	BHFHDI	Housing assets to income (ORIG)	RBA	Q	5
333	BSPNSHUDS	Household dwelling stock (ORIG)	RBA	Q	5
334	BSPNSHUDSYP	Year-ended household dwelling stock growth (ORIG)	RBA	Q	5
374	FILRHL3YF	Lending rates; Housing loans; Banks; 3-year fixed; Owner-occupier (ORIG)	RBA	M	5
375	FILRHL3YFI	Lending rates; Housing loans; Banks; 3-year fixed; Investor (ORIG)	RBA	M	6
376	FILRHLBVD	Lending rates; Housing loans; Banks; Variable; Discounted; Owner-occupier (ORIG)	RBA	M	5
377	FILRHLBVDI	Lending rates; Housing loans; Banks; Variable; Discounted; Investor (ORIG)	RBA	M	5

(continued)

ID	Series ID	Description	Source	Freq	Tcode
378	FILRHLBVDO	Lending rates; Housing loans; Banks; Variable; Standard interest-only; Investor (ORIG)	RBA	M	6
379	FILRHLBVO	Lending rates; Housing loans; Banks; Variable; Standard interest-only; Owner-occupier (ORIG)	RBA	M	6
380	FILRHLBVS	Lending rates; Housing loans; Banks; Variable; Standard; Owner-occupier (ORIG)	RBA	M	5
381	FILRHLBVSI	Lending rates; Housing loans; Banks; Variable; Standard; Investor (ORIG)	RBA	M	6
459	GISPSNBA	Private non-residential building approvals (ORIG)	RBA	M	1

Wages (14 series)

ID	Series ID	Description	Source	Freq	Tcode
260	A2433068T	Unit labour cost - Nominal ; (SA)	ABS	Q	5
261	A2433071F	Unit labour cost - Real ; (SA)	ABS	Q	1
262	A2603039T	Quarterly Index ; Total hourly rates of pay excluding bonuses ; Australia ; Private ; All industries ; (ORIG)	ABS	Q	1
263	A2603609J	Quarterly Index ; Total hourly rates of pay excluding bonuses ; Australia ; Private and Public ; All industries ; (ORIG)	ABS	Q	1
264	A2603989W	Quarterly Index ; Total hourly rates of pay excluding bonuses ; Australia ; Public ; All industries ; (ORIG)	ABS	Q	6
265	A2615009A	Quarterly Index ; Total hourly rates of pay including bonuses ; Australia ; Private ; All industries ; (ORIG)	ABS	Q	6
266	A2615579C	Quarterly Index ; Total hourly rates of pay including bonuses ; Australia ; Private and Public ; All industries ; (ORIG)	ABS	Q	6
267	A2615959F	Quarterly Index ; Total hourly rates of pay including bonuses ; Australia ; Public ; All industries ; (ORIG)	ABS	Q	6
268	A2626979V	Quarterly Index ; Ordinary time hourly rates of pay including bonuses ; Australia ; Private ; All industries ; (ORIG)	ABS	Q	6
269	A2627549L	Quarterly Index ; Ordinary time hourly rates of pay including bonuses ; Australia ; Private and Public ; All industries ; (ORIG)	ABS	Q	6

(continued)

ID	Series ID	Description	Source	Freq	Tcode
270	A2627929R	Quarterly Index ; Ordinary time hourly rates of pay including bonuses ; Australia ; Public ; All industries ; (ORIG)	ABS	Q	6
271	A2713846W	Quarterly Index ; Total hourly rates of pay excluding bonuses ; Australia ; Private ; All industries ; (SA)	ABS	Q	6
272	A2713849C	Quarterly Index ; Total hourly rates of pay excluding bonuses ; Australia ; Private and Public ; All industries ; (SA)	ABS	Q	6
273	A2713852T	Quarterly Index ; Total hourly rates of pay excluding bonuses ; Australia ; Public ; All industries ; (SA)	ABS	Q	6

Commodities (13 series)

ID	Series ID	Description	Source	Freq	Tcode
454	GC=F	Gold price (ORIG)	Yahoo	M	5
466	GRCPAIAD	Commodity prices <U+2013> A$ (ORIG)	RBA	M	5
467	GRCPAISAD	Commodity prices (with bulk commodities spot prices) <U+2013> A$ (ORIG)	RBA	M	5
468	GRCPBCAD	Bulk commodities prices <U+2013> A$ (ORIG)	RBA	M	5
469	GRCPBCSAD	Bulk commodities spot prices <U+2013> A$ (ORIG)	RBA	M	5
470	GRCPBMAD	Base metals prices <U+2013> A$ (ORIG)	RBA	M	5
471	GRCPNRAD	Non-rural commodity prices <U+2013> A$ (ORIG)	RBA	M	5
472	GRCPRCAD	Rural commodity prices <U+2013> A$ (ORIG)	RBA	M	1
475	MCOILBRENTEU	Crude Oil Prices: Brent - Europe (ORIG)	FRED	M	5
476	PCOALAUUSDM	Global price of Coal, Australia (ORIG)	FRED	M	1
477	PIORECRUSDM	Global price of Iron Ore (ORIG)	FRED	M	5
478	POILBREUSDM	Global price of Brent Crude (USD) (ORIG)	FRED	M	5
479	WTISPLC	Spot Crude Oil Price: West Texas Intermediate (WTI) (ORIG)	FRED	M	5

International (13 series)

ID	Series ID	Description	Source	Freq	Tcode
1	000001.SS	Shanghai SSE Composite Index (ORIG)	Yahoo	M	1

(continued)

ID	Series ID	Description	Source	Freq	Tcode
418	FOOIRATCR	Australia Target Cash Rate (ORIG)	RBA	M	1
419	FOOIRCTR	Canada Target Rate (ORIG)	RBA	M	5
420	FOOIREARR	Euro Area Refinancing Rate (ORIG)	RBA	M	2
421	FOOIRJTCR	Japan Policy Rate (ORIG)	RBA	M	2
422	FOOIRUKOBR	United Kingdom Bank Rate (ORIG)	RBA	M	5
423	FOOIRUSFFTRMX	United States Federal Funds Maximum Target Rate (ORIG)	RBA	M	5
482	^FTSE	UK FTSE 100 Index (ORIG)	Yahoo	M	5
483	^GSPC	US S&P 500 (ORIG)	Yahoo	M	5
484	^HSI	Hong Kong Hang Seng Index (ORIG)	Yahoo	M	5
485	^KS11	Korea KOSPI Composite Index (ORIG)	Yahoo	M	1
486	^N100	EU Euronext 100 Index (ORIG)	Yahoo	M	5
487	^N225	Japan Nikkei 225 (ORIG)	Yahoo	M	5

Industrial production and sales (8 series)

ID	Series ID	Description	Source	Freq	Tcode
43	A2298671X	Total industrial industries ; (SA)	ABS	Q	5
56	A2302599C	Domestic sales: Current prices ; (SA)	ABS	Q	5
57	A2302600A	Total sales: Current prices ; (SA)	ABS	Q	5
283	A2716600A	Mining (B) ; Mining excluding exploration and mining support services ; (SA)	ABS	Q	5
284	A2716601C	Manufacturing (C) ; (SA)	ABS	Q	5
285	A2716610F	Electricity, gas, water and waste services (D) ; (SA)	ABS	Q	5
456	GICNBC	Business conditions (SA)	RBA	M	1
460	GISSRTC	Retail sales (SA)	RBA	M	5

Unemployment rate (7 series)

ID	Series ID	Description	Source	Freq	Tcode
298	A84423044F	> Unemployed looked for full-time work ; Persons ; (SA)	ABS	M	5
299	A84423045J	> Unemployed looked for only part-time work ; Persons ; (SA)	ABS	M	5

(continued)

ID	Series ID	Description	Source	Freq	Tcode
300	A84423046K	Unemployed total ; Persons ; (SA)	ABS	M	5
302	A84423050A	Unemployment rate ; Persons ; (SA)	ABS	M	1
304	A84423052F	> Unemployment rate looked for full-time work ; Persons ; (SA)	ABS	M	5
305	A84423053J	> Unemployment rate looked for only part-time work ; Persons ; (SA)	ABS	M	5
320	A85255726K	Underutilisation rate ; Persons ; (SA)	ABS	M	5

Inventories (3 series)

ID	Series ID	Description	Source	Freq	Tcode
55	A2302598A	Private non-farm inventory levels - book values: Current prices ; (SA)	ABS	Q	5
58	A2302601C	Inventories to total sales: Ratio ; (SA)	ABS	Q	5
200	A2304112A	Changes in inventories ; (SA)	ABS	Q	1

Equity indices (2 series)

ID	Series ID	Description	Source	Freq	Tcode
480	^AORD	AUS All Ordinaries (ORIG)	Yahoo	M	1
481	^AXJO	AUS ASX200 (ORIG)	Yahoo	M	1

Expectations and surveys (1 series)

ID	Series ID	Description	Source	Freq	Tcode
457	GICWMICS	Consumer sentiment (SA)	RBA	M	5

Other (1 series)

ID	Series ID	Description	Source	Freq	Tcode
458	GISPSDA	Private dwelling approvals (SA)	RBA	M	1

Appendix C: Detailed Results

RMSFE by Model and Horizon

Table 20: Root mean squared forecast error by model and horizon.

Target	h	RBA	AR(1)	Elastic Net	Historical Mean	LASSO	Random Forest	Ridge	XGBoost
CPI Inflation	0	0.287	0.168	1.021	4.012	0.988	0.542	1.451	0.958
CPI Inflation	1	0.592	0.792	1.185	4.005	1.187	1.274	1.877	1.423
CPI Inflation	2	0.912	1.190	1.243	4.016	1.269	1.240	1.981	1.441
CPI Inflation	4	1.442	1.825	1.684	4.046	1.752	1.432	2.568	1.635
CPI Inflation	8	1.867	2.284	2.225	4.087	2.380	1.482	2.806	1.656
GDP Growth	0	0.797	0.983	0.113	0.958	0.117	0.202	0.202	0.306
GDP Growth	1	1.005	0.977	0.967	0.967	0.967	1.096	2.106	1.153
GDP Growth	2	1.055	0.975	0.970	0.970	0.970	0.991	1.877	1.027
GDP Growth	4	1.066	0.970	0.970	0.970	0.973	0.994	2.047	1.007
GDP Growth	8	1.195	0.983	0.982	0.982	0.982	1.016	1.216	0.997
Underlying Inflation	0	0.289	0.117	0.681	2.168	0.751	0.440	0.943	0.584
Underlying Inflation	1	0.452	0.337	1.045	2.164	1.044	0.996	1.898	0.971
Underlying Inflation	2	0.657	0.566	1.053	2.168	1.063	0.997	1.858	1.073
Underlying Inflation	4	0.986	0.951	1.223	2.179	1.294	1.022	1.542	1.138
Underlying Inflation	8	1.415	1.364	1.763	2.201	1.857	0.929	1.682	1.005
Unemployment Rate	0	0.389	0.045	0.759	1.933	0.720	0.418	1.082	0.553
Unemployment Rate	1	0.555	0.282	0.999	1.936	1.101	0.909	1.421	0.926
Unemployment Rate	2	0.680	0.466	1.093	1.940	1.094	1.019	1.601	1.023
Unemployment Rate	4	0.886	0.742	1.400	1.945	1.356	1.231	2.073	1.297
Unemployment Rate	8	1.075	1.073	1.849	1.997	1.966	1.461	2.294	1.521

MAFE by Model and Horizon

Table 21: Mean absolute forecast error by model and horizon.

Target	h	RBA	AR(1)	Elastic Net	Historical Mean	LASSO	Random Forest	Ridge	XGBoost
CPI Inflation	0	0.209	0.128	0.805	3.682	0.786	0.395	1.081	0.725
CPI Inflation	1	0.430	0.562	0.860	3.671	0.853	0.932	1.393	1.096
CPI Inflation	2	0.659	0.896	0.935	3.684	0.947	0.928	1.511	1.162
CPI Inflation	4	1.047	1.394	1.226	3.708	1.296	1.116	1.936	1.304
CPI Inflation	8	1.293	1.667	1.668	3.747	1.737	1.172	2.168	1.336
GDP Growth	0	0.565	0.539	0.060	0.517	0.060	0.062	0.127	0.116
GDP Growth	1	0.591	0.528	0.525	0.525	0.525	0.569	0.973	0.595
GDP Growth	2	0.632	0.534	0.527	0.527	0.527	0.579	0.928	0.585
GDP Growth	4	0.613	0.524	0.521	0.521	0.524	0.564	1.000	0.561
GDP Growth	8	0.621	0.526	0.525	0.525	0.525	0.582	0.882	0.562
Underlying Inflation	0	0.225	0.049	0.487	1.949	0.520	0.293	0.745	0.470
Underlying Inflation	1	0.347	0.235	0.743	1.940	0.764	0.692	1.073	0.713
Underlying Inflation	2	0.455	0.371	0.808	1.946	0.790	0.716	1.064	0.779
Underlying Inflation	4	0.614	0.601	0.917	1.957	0.974	0.815	1.078	0.875
Underlying Inflation	8	0.899	0.870	1.310	1.978	1.409	0.774	1.111	0.848
Unemployment Rate	0	0.209	0.025	0.579	1.732	0.544	0.333	0.846	0.444
Unemployment Rate	1	0.329	0.183	0.796	1.735	0.807	0.728	1.148	0.756
Unemployment Rate	2	0.437	0.321	0.902	1.738	0.903	0.829	1.227	0.842
Unemployment Rate	4	0.606	0.554	1.116	1.744	1.091	1.019	1.387	1.060
Unemployment Rate	8	0.765	0.831	1.492	1.791	1.581	1.230	1.639	1.200

Forecast Bias by Model and Horizon

Table 22: Forecast bias (mean error) by model and horizon. Positive values indicate under-prediction.

Target	h	RBA	AR(1)	Elastic Net	Historical Mean	LASSO	Random Forest	Ridge	XGBoost
CPI Inflation	0	-0.019	-0.112	0.124	-3.546	0.111	-0.147	-0.409	-0.514
CPI Inflation	1	-0.027	-0.090	-0.044	-3.535	-0.049	-0.414	-0.477	-0.755

Table 22: Forecast bias (mean error) by model and horizon. Positive values indicate under-prediction. *(continued)*

Target	h	RBA	AR(1)	Elastic Net	Historical Mean	LASSO	Random Forest	Ridge	XGBoost
CPI Inflation	2	-0.015	-0.074	0.030	-3.546	0.081	-0.530	-0.383	-0.802
CPI Inflation	4	-0.027	-0.044	0.181	-3.566	0.205	-0.586	-0.215	-0.933
CPI Inflation	8	0.179	-0.051	0.977	-3.601	1.276	-0.404	0.049	-0.773
GDP Growth	0	0.022	-0.032	0.011	-0.032	0.019	-0.007	0.028	-0.010
GDP Growth	1	-0.131	-0.029	-0.035	-0.035	-0.035	-0.036	0.149	-0.032
GDP Growth	2	-0.133	-0.030	-0.030	-0.030	-0.030	-0.029	0.213	-0.021
GDP Growth	4	-0.141	-0.024	-0.020	-0.020	-0.032	-0.040	-0.068	-0.042
GDP Growth	8	-0.296	-0.049	-0.056	-0.056	-0.056	-0.084	0.118	-0.090
Underlying Inflation	0	0.049	-0.003	0.323	-1.703	0.351	-0.094	-0.528	-0.295
Underlying Inflation	1	0.074	0.007	0.324	-1.693	0.369	-0.259	-0.721	-0.332
Underlying Inflation	2	0.089	0.017	0.326	-1.695	0.346	-0.335	-0.657	-0.375
Underlying Inflation	4	0.112	0.037	0.333	-1.701	0.402	-0.363	-0.459	-0.404
Underlying Inflation	8	0.188	0.079	0.897	-1.717	1.032	-0.350	-0.341	-0.527
Unemployment Rate	0	-0.157	-0.014	-0.318	-1.369	-0.274	-0.245	-0.647	-0.358
Unemployment Rate	1	-0.238	-0.065	-0.513	-1.412	-0.494	-0.562	-0.873	-0.571
Unemployment Rate	2	-0.295	-0.116	-0.571	-1.455	-0.531	-0.639	-0.926	-0.661
Unemployment Rate	4	-0.331	-0.222	-0.804	-1.544	-0.696	-0.779	-0.965	-0.873
Unemployment Rate	8	-0.219	-0.406	-1.347	-1.692	-1.121	-0.914	-1.057	-1.000

Relative RMSFE (ML / RBA)

Table 23: Relative RMSFE (ML model / RBA). Values below 1.0 indicate ML outperformance.

Target	h	AR(1)	Elastic Net	Historical Mean	LASSO	Random Forest	Ridge	XGBoost
CPI Inflation	0	0.586	3.555	13.969	3.441	1.889	5.052	3.337
CPI Inflation	1	1.339	2.003	6.766	2.005	2.152	3.171	2.404
CPI Inflation	2	1.305	1.364	4.406	1.393	1.361	2.173	1.581
CPI Inflation	4	1.266	1.168	2.806	1.215	0.993	1.781	1.134

Table 23: Relative RMSFE (ML model / RBA). Values below 1.0 indicate ML outperformance. *(continued)*

Target	h	AR(1)	Elastic Net	Historical Mean	LASSO	Random Forest	Ridge	XGBoost
CPI Inflation	8	1.223	1.192	2.189	1.275	0.794	1.503	0.887
GDP Growth	0	1.233	0.142	1.201	0.147	0.253	0.253	0.384
GDP Growth	1	0.971	0.962	0.962	0.962	1.090	2.095	1.147
GDP Growth	2	0.924	0.919	0.919	0.919	0.939	1.778	0.973
GDP Growth	4	0.910	0.911	0.911	0.913	0.933	1.921	0.945
GDP Growth	8	0.822	0.821	0.821	0.821	0.850	1.017	0.834
Underlying Inflation	0	0.406	2.352	7.488	2.595	1.519	3.259	2.019
Underlying Inflation	1	0.745	2.313	4.789	2.311	2.204	4.200	2.149
Underlying Inflation	2	0.860	1.601	3.298	1.618	1.517	2.826	1.632
Underlying Inflation	4	0.965	1.241	2.211	1.313	1.037	1.565	1.155
Underlying Inflation	8	0.964	1.246	1.555	1.312	0.657	1.189	0.711
Unemployment Rate	0	0.115	1.950	4.966	1.849	1.073	2.781	1.421
Unemployment Rate	1	0.508	1.801	3.490	1.985	1.638	2.561	1.670
Unemployment Rate	2	0.685	1.606	2.851	1.608	1.498	2.353	1.504
Unemployment Rate	4	0.837	1.581	2.196	1.531	1.390	2.341	1.464
Unemployment Rate	8	0.998	1.719	1.857	1.828	1.359	2.133	1.414

Diebold–Mariano Test Results

Table 24: Diebold–Mariano test results. Positive DM indicates ML outperformance. HAC standard errors with $h-1$ lags. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Target	h	Model	DM	p-value	Sig.	N
CPI Inflation	0	AR(1)	3.06	0.002	***	124
CPI Inflation	0	Elastic Net	-7.18	0.000	***	124
CPI Inflation	0	Historical Mean	-14.12	0.000	***	124
CPI Inflation	0	LASSO	-7.30	0.000	***	124
CPI Inflation	0	Random Forest	-4.33	0.000	***	124
CPI Inflation	0	Ridge	-5.85	0.000	***	124

Table 24: Diebold–Mariano test results. Positive DM indicates ML outperformance. HAC standard errors with $h-1$ lags. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. *(continued)*

Target	h	Model	DM	p-value	Sig.	N
CPI Inflation	0	XGBoost	-4.79	0.000	***	124
CPI Inflation	1	AR(1)	-1.89	0.059	*	123
CPI Inflation	1	Elastic Net	-3.99	0.000	***	123
CPI Inflation	1	Historical Mean	-13.76	0.000	***	123
CPI Inflation	1	LASSO	-3.73	0.000	***	123
CPI Inflation	1	Random Forest	-4.69	0.000	***	123
CPI Inflation	1	Ridge	-5.35	0.000	***	123
CPI Inflation	1	XGBoost	-5.46	0.000	***	123
CPI Inflation	2	AR(1)	-2.12	0.034	**	123
CPI Inflation	2	Elastic Net	-1.79	0.074	*	123
CPI Inflation	2	Historical Mean	-7.80	0.000	***	123
CPI Inflation	2	LASSO	-1.87	0.061	*	123
CPI Inflation	2	Random Forest	-1.65	0.099	*	123
CPI Inflation	2	Ridge	-3.69	0.000	***	123
CPI Inflation	2	XGBoost	-2.75	0.006	***	123
CPI Inflation	4	AR(1)	-1.51	0.132		122
CPI Inflation	4	Elastic Net	-0.90	0.367		122
CPI Inflation	4	Historical Mean	-4.68	0.000	***	122
CPI Inflation	4	LASSO	-1.10	0.270		122
CPI Inflation	4	Random Forest	-0.05	0.964		122
CPI Inflation	4	Ridge	-2.84	0.005	***	122
CPI Inflation	4	XGBoost	-0.63	0.530		122
CPI Inflation	8	AR(1)	-2.25	0.024	**	72
CPI Inflation	8	Elastic Net	-1.28	0.200		72
CPI Inflation	8	Historical Mean	-2.29	0.022	**	72
CPI Inflation	8	LASSO	-1.43	0.154		72
CPI Inflation	8	Random Forest	0.80	0.426		72
CPI Inflation	8	Ridge	-1.57	0.117		72
CPI Inflation	8	XGBoost	0.56	0.576		72
GDP Growth	0	AR(1)	-0.88	0.378		125
GDP Growth	0	Elastic Net	5.03	0.000	***	125

Table 24: Diebold–Mariano test results. Positive DM indicates ML outperformance. HAC standard errors with $h-1$ lags. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. *(continued)*

Target	h	Model	DM	p-value	Sig.	N
GDP Growth	0	Historical Mean	-0.76	0.449		125
GDP Growth	0	LASSO	5.03	0.000	***	125
GDP Growth	0	Random Forest	4.89	0.000	***	125
GDP Growth	0	Ridge	4.86	0.000	***	125
GDP Growth	0	XGBoost	4.17	0.000	***	125
GDP Growth	1	AR(1)	0.10	0.924		124
GDP Growth	1	Elastic Net	0.30	0.765		124
GDP Growth	1	Historical Mean	0.30	0.765		124
GDP Growth	1	LASSO	0.30	0.765		124
GDP Growth	1	Random Forest	-0.86	0.387		124
GDP Growth	1	Ridge	-1.34	0.180		124
GDP Growth	1	XGBoost	-1.06	0.290		124
GDP Growth	2	AR(1)	0.89	0.375		124
GDP Growth	2	Elastic Net	0.97	0.330		124
GDP Growth	2	Historical Mean	0.97	0.330		124
GDP Growth	2	LASSO	0.97	0.330		124
GDP Growth	2	Random Forest	0.81	0.415		124
GDP Growth	2	Ridge	-1.15	0.249		124
GDP Growth	2	XGBoost	0.11	0.913		124
GDP Growth	4	AR(1)	1.30	0.192		122
GDP Growth	4	Elastic Net	1.29	0.197		122
GDP Growth	4	Historical Mean	1.29	0.197		122
GDP Growth	4	LASSO	1.25	0.211		122
GDP Growth	4	Random Forest	0.89	0.375		122
GDP Growth	4	Ridge	-1.24	0.216		122
GDP Growth	4	XGBoost	0.69	0.489		122
GDP Growth	8	AR(1)	1.48	0.139		77
GDP Growth	8	Elastic Net	1.54	0.124		77
GDP Growth	8	Historical Mean	1.54	0.124		77
GDP Growth	8	LASSO	1.54	0.124		77
GDP Growth	8	Random Forest	0.69	0.488		77

Table 24: Diebold–Mariano test results. Positive DM indicates ML outperformance. HAC standard errors with $h-1$ lags. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. *(continued)*

Target	h	Model	DM	p-value	Sig.	N
GDP Growth	8	Ridge	-2.78	0.005	***	77
GDP Growth	8	XGBoost	0.90	0.368		77
Underlying Inflation	0	AR(1)	5.29	0.000	***	124
Underlying Inflation	0	Elastic Net	-4.11	0.000	***	124
Underlying Inflation	0	Historical Mean	-12.68	0.000	***	124
Underlying Inflation	0	LASSO	-4.51	0.000	***	124
Underlying Inflation	0	Random Forest	-1.75	0.081	*	124
Underlying Inflation	0	Ridge	-6.73	0.000	***	124
Underlying Inflation	0	XGBoost	-5.64	0.000	***	124
Underlying Inflation	1	AR(1)	4.05	0.000	***	123
Underlying Inflation	1	Elastic Net	-3.26	0.001	***	123
Underlying Inflation	1	Historical Mean	-12.28	0.000	***	123
Underlying Inflation	1	LASSO	-3.85	0.000	***	123
Underlying Inflation	1	Random Forest	-3.86	0.000	***	123
Underlying Inflation	1	Ridge	-1.66	0.097	*	123
Underlying Inflation	1	XGBoost	-3.93	0.000	***	123
Underlying Inflation	2	AR(1)	1.97	0.049	**	123
Underlying Inflation	2	Elastic Net	-2.74	0.006	***	123
Underlying Inflation	2	Historical Mean	-6.86	0.000	***	123
Underlying Inflation	2	LASSO	-2.51	0.012	**	123
Underlying Inflation	2	Random Forest	-1.93	0.053	*	123
Underlying Inflation	2	Ridge	-1.50	0.133		123
Underlying Inflation	2	XGBoost	-1.93	0.053	*	123
Underlying Inflation	4	AR(1)	0.75	0.454		122
Underlying Inflation	4	Elastic Net	-1.16	0.246		122
Underlying Inflation	4	Historical Mean	-3.61	0.000	***	122
Underlying Inflation	4	LASSO	-1.48	0.139		122
Underlying Inflation	4	Random Forest	-0.18	0.861		122
Underlying Inflation	4	Ridge	-1.80	0.071	*	122
Underlying Inflation	4	XGBoost	-0.87	0.386		122
Underlying Inflation	8	AR(1)	-1.32	0.188		74

Table 24: Diebold–Mariano test results. Positive DM indicates ML outperformance. HAC standard errors with $h-1$ lags. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. *(continued)*

Target	h	Model	DM	p-value	Sig.	N
Underlying Inflation	8	Elastic Net	-1.12	0.262		74
Underlying Inflation	8	Historical Mean	-1.24	0.215		74
Underlying Inflation	8	LASSO	-1.20	0.232		74
Underlying Inflation	8	Random Forest	0.87	0.382		74
Underlying Inflation	8	Ridge	-1.64	0.101		74
Underlying Inflation	8	XGBoost	0.71	0.480		74
Unemployment Rate	0	AR(1)	1.82	0.069	*	126
Unemployment Rate	0	Elastic Net	-3.91	0.000	***	126
Unemployment Rate	0	Historical Mean	-11.64	0.000	***	126
Unemployment Rate	0	LASSO	-3.54	0.000	***	126
Unemployment Rate	0	Random Forest	-0.26	0.794		126
Unemployment Rate	0	Ridge	-6.66	0.000	***	126
Unemployment Rate	0	XGBoost	-1.64	0.100		126
Unemployment Rate	1	AR(1)	2.21	0.027	**	125
Unemployment Rate	1	Elastic Net	-5.41	0.000	***	125
Unemployment Rate	1	Historical Mean	-10.54	0.000	***	125
Unemployment Rate	1	LASSO	-3.01	0.003	***	125
Unemployment Rate	1	Random Forest	-3.79	0.000	***	125
Unemployment Rate	1	Ridge	-7.85	0.000	***	125
Unemployment Rate	1	XGBoost	-3.97	0.000	***	125
Unemployment Rate	2	AR(1)	1.47	0.141		125
Unemployment Rate	2	Elastic Net	-3.83	0.000	***	125
Unemployment Rate	2	Historical Mean	-5.88	0.000	***	125
Unemployment Rate	2	LASSO	-3.74	0.000	***	125
Unemployment Rate	2	Random Forest	-2.39	0.017	**	125
Unemployment Rate	2	Ridge	-4.74	0.000	***	125
Unemployment Rate	2	XGBoost	-2.57	0.010	**	125
Unemployment Rate	4	AR(1)	1.31	0.192		123
Unemployment Rate	4	Elastic Net	-2.38	0.017	**	123
Unemployment Rate	4	Historical Mean	-3.74	0.000	***	123
Unemployment Rate	4	LASSO	-2.80	0.005	***	123

Table 24: Diebold–Mariano test results. Positive DM indicates ML outperformance. HAC standard errors with $h-1$ lags. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. *(continued)*

Target	h	Model	DM	p-value	Sig.	N
Unemployment Rate	4	Random Forest	-1.90	0.058	*	123
Unemployment Rate	4	Ridge	-1.95	0.052	*	123
Unemployment Rate	4	XGBoost	-2.26	0.024	**	123
Unemployment Rate	8	AR(1)	0.27	0.788		75
Unemployment Rate	8	Elastic Net	-0.83	0.405		75
Unemployment Rate	8	Historical Mean	-3.48	0.001	***	75
Unemployment Rate	8	LASSO	-0.83	0.409		75
Unemployment Rate	8	Random Forest	-0.84	0.402		75
Unemployment Rate	8	Ridge	-1.50	0.134		75
Unemployment Rate	8	XGBoost	-0.96	0.335		75

Appendix D: RBA Forecast Publication History

The RBA's macroeconomic forecasts have been published in the *Statement on Monetary Policy* (SMP) since 1997. Prior to this, forecasts were included in various RBA publications and internal documents.

From 1997 to 2006, the SMP was published semi-annually, typically in May and November. From February 2007, publication frequency increased to quarterly (February, May, August, and November). The format of the forecast tables has evolved over time, with the current format reporting year-ended percentage changes for inflation variables and quarter-on-quarter changes for GDP growth.

The RBA's forecasting methodology has also developed over the evaluation period. In the 1990s and early 2000s, forecasts relied primarily on structural models and staff judgement. More recently, the Bank has incorporated a wider range of statistical models, including dynamic stochastic general equilibrium (DSGE) models and factor models, alongside its core forecasting framework. The 2023 independent Review of the Reserve Bank (RBA Review Panel, 2023) recommended further enhancements to the Bank's analytical toolkit and forecast communication.

Our analysis uses forecast data compiled by the `readrba` R package (Cowgill, 2023b), which extracts historical forecast tables from published editions of the SMP. The forecast origins in our evaluation sample span 1993 Q1 to 2025 Q4, with the number of available forecast horizons varying by origin date. The 1993 Q1 start date corresponds to the formal adoption of the 2–3 per cent inflation target.

Appendix E: Replication

All code and data required to replicate the analysis in this paper are available at github.com/andybridger/RBAvsMachine. The pipeline is structured as follows:

Table 25: Replication pipeline.

Script	Description
`R/00_setup.R`	Package installation and environment configuration
`R/01_data_pull.R`	Pull data from ABS, RBA, FRED, and Yahoo Finance
`R/02_stationarity.R`	ADF-based transformation selection and application
`R/03_rba_forecasts.R`	RBA forecast extraction and error computation
`R/04_ml_models.R`	Expanding-window ML model training
`R/05_evaluation.R`	Forecast comparison and statistical tests
`R/06_exhibits.R`	Figure and table generation

Data series identifiers are stored in CSV files under `data/config/`, allowing the database to be updated by re-running the pipeline. Raw data are saved in compressed RDS format; the full panel occupies approximately 2 MB.

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